

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING**EC3452 -ELECTROMAGNETIC FIELDS****CLASS/SEM - II ECE/IV SEM****UNIT I - INTRODUCTION****Part A****1. Define scalar field?**

A field is a system in which a particular physical function has a value at each and every point in that region. The distribution of a scalar quantity with a defined position in a space is called scalar field.

Ex: Temperature of atmosphere.

2. Define Vector field?

If a quantity which is specified in a region to define a field is a vector then the corresponding field is called vector field.

3. Define scaling of a vector?

This is nothing but, multiplication of a scalar with a vector. Such a multiplication changes the magnitude of a vector but not the direction.

4. What are co-planar vector?

The vectors which lie in the same plane are called co-planar vectors.

5. Define base vectors?

The base vectors are the unit vectors which are strictly oriented along the directions of the coordinate axes of the given coordinate system.

6. What is a position vector?

Consider a point $p(x, y, z)$ are Cartesian coordinate system. Then the position vector of point p is represented by the distance of point p from the origin directed from origin to point. This is also called as radius vector.

7. Define Divergence.

Divergence is defined as the net outward flow of the flux per unit volume over a closed incremental surface.

8. State Divergence Theorem.

The integral of the normal component of any vector field over a closed surface is equal to the integral of the divergence of this vector field throughout the volume enclosed that closed surface.

9. What is physical significance of curl of a vector field?

Curl gives rate of rotation. $\text{Curl } \mathbf{F}$ gives work done per unit area.

10. What is physical significance of divergence?

Divergence of current density gives net outflow of current per unit volume. Divergence of flux density gives net outflow per unit volume. In general, divergence of any field density gives net outflow of that field per unit volume.

11. State the conditions for a field to be a) solenoid b) irrotational.

a) Divergence of the field has to be zero.

b) Curl of the field has to be zero.

12. Define scalar and vector quantity?

The scalar is a quantity whose value may be represented by a single real number which may be positive or negative. e.g, temperature, mass, volume, density.

A quantity which has both a magnitude and a specified direction in space is called a vector. e.g. force, velocity, displacement, acceleration.

13. What is a unit vector? What is its function while representing a vector?

A unit vector has a function to indicate the direction. Its magnitude is always unity, irrespective of the direction which it indicates and the coordinate system under consideration.

14. Name 3 coordinate systems used in electromagnetic engineering?

1) Cartesian or rectangular coordinate system.

2) Cylindrical coordinate system.

3) Spherical coordinate system.

15. How to represent a point in a Cartesian system?

A point in rectangular coordinate system is located by three coordinates namely x, y and z coordinates. The point can be reached by moving from origin, the distance x in x direction then the distance y in y direction and finally z in z direction.

16. What is separation of vector?

The distance vector is also called as separation vector. Distance vector is nothing but the length of the vector.

17. State Distance formula?

Distance formula give the distance between the two points representing tips of the vector.

18. Show how a point p represented in a spherical coordinate system.

The point p can be defined as the intersection of three surfaces in spherical coordinate System.

r - Constant which is a sphere with centre as origin

θ - Constant which is a right circular cone with apex as origin and axis as z axis.

Φ - Constant is a plane perpendicular to xy plane.

19. What are the types of integral related to electromagnetic theory?

1. Line integral
2. Surface integral
3. Volume integral

20. Give the types of charge distribution.

1. Line charge
2. Point charge
3. Surface charge
4. Volume charge.

Part B

1. (a) State and prove divergence theorem.

(b) What are the major sources of electromagnetic fields?

2. Check validity of the divergence theorem considering the field $D = 2xy \hat{a}_x + x^2 \hat{a}_y$ c/m² and the rectangular parallelepiped formed by the planes $x=0, x=1, y=0, y=2$ & $z=0, z=3$.

3. A vector field $D = [5r^2/4]\hat{r}$ is given in spherical co-ordinates. Evaluate both sides of divergence theorem for the volume enclosed between $r=1$ & $r=2$.

4. Explain three co-ordinate systems.
5. State and prove Gauss law and explain applications of Gauss law.
6. Derive an expression for the electric field due to a straight and infinite uniformly charged wire of length 'L' meters and with a charge density of a point P which lies along the perpendicular bisector of wire.
7. A circular disc of radius 'a' m is charged uniformly with a charge density. find the electric field at a point 'h' m from the disc along its axis.
8. Define the potential difference and absolute potential. Give the relation between potential and field intensity.
9. Derive an expression for potential due to infinite uniformly charged line and also derive potential due to electric dipole.

UNIT II – ELECTROSTATICS

Part A

1. Define point charge.

A point charge means that electric charge which is separated on a surface or space whose geometrical dimensions are very small compared to other dimensions, in which the effect of electric field to be studied.

2. Define one coulomb.

One coulomb of charge is defined as the charge possessed by $(1/1.602 \times 10^{-19})$ i.e 6×10^{18} number of electrons.

3. What are the various types of charge distribution? Give an example for each.

1. Point charge - Ex. Positive charge
2. Line charge - Ex. A sharp beam in a cathode ray tube.
3. Surface charge - Ex. The plate of a charged parallel plate capacitor.
4. Volume charge - Ex. The charged cloud.

4. State the assumptions made while defining a Coulomb's law.

- 1) The two charges are stationary.
- 2) The two charges are point charge.

5. What is an equipotential surface?

An equipotential surface is an imaginary surface in an electric field of a given charge distribution, in which all points on the surface are at the same electric potential.

6. What is an electric flux?

The total number of lines of force in any particular electric field is called electric flux. It is represented by the symbol ϕ . Similar to the charge, unit of electric flux is also Coulomb.

7. Define electric flux density.

The net flux passing normal through the unit surface area is called electric flux density. It is denoted as D . It has a specified direction which is normal to the surface area under consideration hence it is a vector field.

8. State Gauss's Law.

The electric flux passing through any closed surface is equal to the total charge enclosed by that surface.

9. State the application of Gauss's law.

- 1) The Gauss's law can be used to find E and D for symmetrical charge distributions.
- 2) It is used to find the charge enclosed or the flux passing through the closed surface.

10. What is an equipotential surface?

An equipotential surface is an imaginary surface in an electric field of a given charge distribution, in which all points on the surface are at the same electric potential.

11. Define the unit of Potential difference.

The unit of potential difference is Volt. One Volt potential difference is one Joule of work done in moving unit charge from one point to other in the field.

12. Define potential difference.

The work done per unit charge in moving unit charge from B to A in the field E is called potential difference between the points B to A.

13. Define relaxation time.

The relaxation time δ is defined as the time required by the charge density to decay to 36.8% of its initial value.

14. What is Potential Gradient?

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The rate of change of potential with respect to the distance is called potential gradient

15. What is Gaussian surface? What are the conditions to be satisfied in special Gaussian surface?

The surface over which the Gauss's law is applied is called Gaussian surface. Obviously such a surface is a closed surface and it has to satisfy the following conditions.

- 1) The surface may be irregular but should be sufficiently large so as to enclose the entire charge.
- 2) The surface must be closed.
- 3) At each point of the surface D is either normal or tangential to the surface.
- 4) The electric flux density D is constant over the surface at which D is normal.

16. What is Gradient of V ?

The maximum value of rate of change of potential with distance dv/dL is called gradient of V

17. Define Absolute potential.

The work done in moving a unit charge from infinity (or from reference point at which potential is zero) to the point under the consideration against E is called absolute potential of that point.

18. What is Polarization?

The applied field E shifts the charges inside the dielectric to induce the electric dipoles. This process is called Polarization.

19. What is Polarization of Dielectrics?

Polarization of dielectric means, when an electron cloud has a centre separated from the nucleus. This forms an electric dipole. The dipole gets aligned with the applied field.

20. What is method of images?

The replacement of the actual problem with boundaries by an enlarged region or with image charges but no boundaries is called the method of images.

21. When is method of images used?

Method of images is used in solving problems of one or more point charges in the presence of boundary surfaces.

Part B

1. Derive the boundary conditions of the normal and tangential components of electric field at the interface of two media with different dielectrics.

2. Derive an expression for the capacitance of a parallel plate capacitor having two dielectric media.
3. Prove Laplace's and Poisson's equations.
4. Derive an expression for the capacitance of two wire transmission line.
5. Briefly explain about the application of Poisson's and Laplace's equations.
6. Derive the expression for co-efficient of coupling.
7. Briefly explain about the wave incident
 - (i) Normally on perfect conductor
 - (ii) Obliquely to the surface of perfect conductor.
8. Derive an expression for capacitance of co-axial cable.

UNIT III – MAGNETOSTATICS

Part A

1. What is Magnetic Field?

The region around a magnet within which influence of the magnet can be experienced is called Magnetic Field.

2. What are Magnetic Lines of Force?

The existence of Magnetic Field can be experienced with the help of compass field. Such a field is represented by imaginary lines around the magnet which are called Magnetic Lines of Force.

3. State Stoke's Theorem.

The line integral of F around a closed path L is equal to the integral of curl of F over the open surface S enclosed by the closed path L .

4. Define scalar magnetic Potential.

The scalar magnetic potential V_m can be defined for source free region where J i.e. current density is zero.

5. What is the fundamental difference between static electric and magnetic field lines?

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There is a fundamental difference between static electric and magnetic field lines. The tubes of electric flux originate and terminate on charges, whereas magnetic flux tubes are continuous.

6. State Kirchoff's Flux law.

It states that the total magnetic flux arriving at any junction in a magnetic circuit is equal to the magnetic flux leaving that junction. Using this law, parallel magnetic circuits can be easily analyzed.

7. State Kirchoff's MMF law.

Kirchoff's MMF law states that the resultant MMF around a closed magnetic circuit is equal to the algebraic sum of products of flux and reluctance of each part of the closed circuit.

8. What is Magnetization?

The field produced due to the movement of bound charges is called Magnetization represented by M .

9. State Biot Savart Law.

The Biot Savart law states that, the magnetic field intensity dH produced at a point p due to a differential current element IdL is,

- 1) Proportional to the product of the current I and differential length dL .
- 2) The sine of the angle between the element and the line joining point p to the element and
- 3) Inversely proportional to the square of the distance R between point p and the element.

$$dH \propto Idl \sin\theta/R^2$$

10. Describe what are the sources of electric field and magnetic field?

Stationary charges produce electric field that are constant in time, hence the term electrostatics. Moving charges produce magnetic fields hence the term magnetostatics.

11. Define Magnetic flux density.

The total magnetic lines of force i.e. magnetic flux crossing a unit area in a plane at right angles to the direction of flux is called magnetic flux density. Unit is Wb/m^2 .

12. State Ampere's circuital law.

The line integral of magnetic field intensity H around a closed path is exactly equal to the direct current enclosed by that path.

13. Define Magnetic field Intensity.

Magnetic Field intensity at any point in the magnetic field is defined as the force experienced

by a unit north pole of one Weber strength, when placed at that point. Unit is N/Wb.

14. What is rotational and irrotational vector field?

If curl of a vector field exists then the field is called rotational. For irrotational vector field, the curl vanishes i.e. curl is zero.

15. Give the application of Stoke's theorem.

The Stoke's theorem is applicable for the open surface enclosed by the given closed path. Any volume is a closed surface and hence application of Stoke's theorem to a closed surface which enclosed certain volume produces zero answer.

Part B

1. Derive the expression for magnetic field intensity and magnetic flux density due to finite and infinite line.
2. Derive the expressions for magnetic field intensity and magnetic flux density due to circular coil.
3. Derive an expression for force between two current carrying conductors.
4. State Ampere's circuital law and explain any two applications of Ampere's circuital law.
5. Derive the expression for the magnetic field intensity due to rectangular coil carrying current I in a uniform field. Deduce the equation to find the H due to square coil.
6. State Ampere's circuital law and prove the same.
7. Find the magnetic field intensity at the centre O of a square loop of sides equal to $5M$ and carrying $10A$ of current.
8. An iron ring with a cross sectional area of $3cm$ square and mean circumference of $15 cm$ is wound with 250 turns wire carrying a current of $0.3A$. The relative permeability of ring is 1500 . Calculate the flux established in the ring.

the form of a ring and thereby closed on itself it becomes a toroid.

UNIT IV – TIME VARYING FIELDS AND MAXWELL'S EQUATIONS

Part A

1. Define a wave.

If a physical phenomenon that occurs at one place at a given time is reproduced at other places at later times, the time delay being proportional to the space separation from the first location then the group of phenomena constitutes a wave.

2. Mention the properties of uniform plane wave.

- i) At every point in space, the electric field E and magnetic field H are perpendicular to each other.
- ii) The fields vary harmonically with time and at the same frequency everywhere in space.

3. Define intrinsic impedance or characteristic impedance.

It is the ratio of electric field to magnetic field or it is the ratio of square root of permeability to permittivity of medium.

4. Give the characteristic impedance of free space.

377ohms.

5. Define skin depth.

It is defined as that depth in which the wave has been attenuated to $1/e$ or approximately 37% of its original value.

6. Define Poynting vector.

The pointing vector is defined as rate of flow of energy of a wave as it propagates. $P = E \times H$

7. State Poynting Theorem.

The net power flowing out of a given volume is equal to the time rate of decrease of the energy stored within the volume conduction losses.

8. Give the difficulties in FDM.

FDM is difficult to apply for problems involving irregular boundaries and non-homogeneous material properties.

9. Explain the steps in finite element method.

- i) Discrimination of the solution region into elements.
- ii) Generation of equations for fields at each element.
- iii) Assembly of all elements.
- iv) Solution of the resulting system.

10. State Maxwell's fourth equation.

The net magnetic flux emerging through any closed surface is zero.

11. State Maxwell's Third equation.

The total electric displacement through the surface enclosing a volume is equal to the total charge within the volume.

12. State the principle of superposition of fields.

The total electric field at a point is the algebraic sum of the individual electric field at that point.

13. Define loss tangent.

Loss tangent is the ratio of the magnitude of conduction current density to displacement current density of the medium.

14. Define reflection coefficient.

Reflection coefficient is defined as the ratio of the magnitude of the reflected field to that of the incident field.

15. Define transmission coefficient.

Transmission coefficient is defined as the ratio of the magnitude of the transmitted field to that of incident field.

16. What will happen when the wave is incident obliquely over dielectric – dielectric boundary?

When a plane wave is incident obliquely on the surface of a perfect dielectric part of the energy is transmitted and part of it is reflected. But in this case the transmitted wave will be refracted, that is the direction of propagation is altered.

17. What are uniform plane waves?

Electromagnetic waves which consist of electric and magnetic fields that are perpendicular to each other and to the direction of propagation and are uniform in plane perpendicular to the direction of propagation are known as uniform plane waves.

18. What is the significant feature of wave propagation in an imperfect dielectric?

The only significant feature of wave propagation in an imperfect dielectric compared to that in a perfect dielectric is the attenuation undergone by the wave.

19. What is the major drawback of finite difference method?

The major drawback of finite difference method is its inability to handle curved boundaries accurately.

20. Define power density.

The power density is defined as the ratio of power to unit area. $\text{Power density} = \text{power} / \text{unit area}$.

21. What is Normal Incidence?

When a uniform plane wave incidences normally to the boundary between the media then it is known as normal incidence.

22. What is called attenuation constant?

When a wave propagates in the medium, it gets attenuated. The amplitude of the signal reduces. This is represented by attenuation constant α . It is measured in Neper per meter

(NP/m). But practically it is expressed in decibel (dB).

23. What is phase constant?

When a wave propagates, phase change also takes place. Such a phase change is expressed by a phase constant $\hat{a}$. It is measured in radian per meter (rad/m).

24. How voltage maxima and minima are separated?

In general voltage minima are separated by one half wavelength. Also the voltage maxima are also separated by one half wave length.

Part B

1. With necessary explanation, derive the Maxwell's equation in differential and integral forms.
2. Write short notes on faradays law of electromagnetic induction.
3. The magnetic field intensity in free space is given as $H = H_0 \sin \theta$ yt A/m. Where $\theta = ut - \hat{a}z$ and $\hat{a}$ is a constant quantity. Determine the displacement current density.
4. (a) What is the physical significance of the poynting vector?
(b) State and explain the poynting theorem.
5. Derive the general wave equation.
6. Derive an expression for energy stored and energy density in an Electrostatic field.
7. Derive a wave equation for non dissipative medium making use of Maxwell equations and field vectors E and H.
8. Define wave. Derive the free space electromagnetic wave equation.

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II year IV sem ECE

Two marks with answers

Unit -1

1. Define volume charge density?

Consider a charge distributed uniformly over a volume. If a differential charge element dQ is present in the differential volume element dv , then the volume charge density is defined as

$$\rho_v = \frac{dQ}{dv} \text{ C/m}^3$$

2. State stokes theorem.

The line integral of a vector around a closed path is equal to the surface integral of the normal component of its curl over any surface bounded by the path

3. Define electric field intensity.

Electric field intensity is defined as the electric force per unit positive charge. $E = F/Q$

$$E = \frac{Q}{4\pi\epsilon_0 r^2} \text{ V/m}$$

4. State Divergence Theorem.

The integral of the divergence of a vector over a volume v is equal to the surface integral of the normal component of the vector over the surface bounded by the volume.

5. Define electric scalar potential.

Potential at any point is defined as the work done in moving a unit positive charge from infinity to that point in an electric field.

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

6. State coulombs law.

Coulombs law states that the force between any two point charges is directly proportional to the product of their magnitudes and inversely proportional to the square of the distance between them. It is directed along the line joining the two charges.

7. Show that the two vectors $A=6ax+ay-5az$ and $B= 3(ax-ay+az)$ are perpendicular to each other

$$\begin{aligned} A \cdot B &= 6 \times 3 + 1 \times (-3) + (-5) \times (-3) \\ &= 18 - 3 + 15 \\ &= 30 \end{aligned}$$

8. What are the conditions for two vectors A and B to be a parallel and perpendicular

For parallel

$$A \times B = 0$$

For perpendicular

$$A \cdot B = 0$$

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9. Express the value of differential volume in rectangular and cylindrical co-ordinate system For rectangular co-ordinate

$$dv = dx \, dy \, dz$$

For cylindrical co-ordinates $dv = r \, dr \, d\theta \, dz$

10. Represent point (0,1,1)m given in Cartesian co-ordinates in spherical co-ordinates Cartesian co-ordinates $x=0, y=1m, z=1m$

$$r^2 = x^2 + y^2 + z^2 = 0 + 1 + 1$$

$$+1 \quad r = 2^{1/2} m$$

$$\cos \theta =$$

$$1/\sqrt{2}$$

$$\tan \phi = 1/0 =$$

$$\infty$$

11. State coulombs law

Coulomb stated that the force between two very small charged objects separated by a large distance compared to their size is proportional to the charge on each object and inversely proportional to the square of the distance between them

$$F \propto (Q_1 Q_2)/r^2$$

12. State gauss's law

The electric flux passing through any closed surface is equal to the total charge enclosed by that surface

13. Define dipole and dipole moment

Dipole or electric dipole is nothing but two equal and opposite point charges are separated by a very small distance. The product of electric charge and distance is known as dipole moment. It is denoted by m where Q is the charge and l is the length

14. What is point charge?

Point charge is one whose maximum dimension is very small in comparison with any other length

15. Name few applications of gauss law in electrostatics

Gauss law is applied to determine the electric field intensity from a closed surface e.x: electric field can be determined for shell, two concentric shell or cylinders, etc

16. Define current density

Current density is defined as current per unit area. It is denoted by $J = I/A \text{ amp/m}^2$

Unit-2

1. State Biot –Savarts law.

It states that the magnetic flux density at any point due to current element is proportional to the current element and sine of the angle between the elemental length and inversely proportional to the square of the distance between them

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$$dB = \mu_0 Idl \sin\theta / 4\pi r^2$$

2. Write Lorentz force equation & mention its applications. Lorentz force equation is given by

$$F = ma$$

$$= Q(E + V \times B)$$

N

Applications

- Used to determine electron or bits in magnetron.
- Used to determine the proton path in cyclotron.

3. Mention the importance of Lorentz force equation.

Lorentz force equation relates mechanical force to the electric force. Lorentz force equation is given by

$$F = ma$$

$$= Q(E + V \times B)$$

4. What is Lorentz force equation?

Lorentz force is the force experienced by the test charge. It is maximum if the direction of movement of charge is perpendicular to the orientation of field lines.

Lorentz force equation is given

$$by F = ma = Q(E + V \times B)$$

5. Brief about Ampere's circuital law in integral form.

The line integral of the magnetic field intensity around a closed path is equal to the sum of the currents flowing through the area enclosed by the path.

$$\oint \vec{H} \cdot d\vec{l} = \sum I$$

6. Brief about complex Poynting vector. Complex Poynting vector P is

$$S_{av}(r) = \frac{1}{2} \text{Re}[\underline{E} \times \underline{H}^*]$$

7. Compare diamagnetic, paramagnetic and ferromagnetic materials

Diamagnetic: in diamagnetic materials magnetization is opposed to the applied field. It has weak magnetic field

Paramagnetic field: in paramagnetic materials magnetization is in the same direction as the field. It has weak magnetic field

Ferromagnetic: in ferromagnetic materials magnetization is in the same direction as the field. It has strong magnetic field

8. Write down the magnetic boundary conditions

The normal components of flux density B is continuous across the boundary

The tangential component of field intensity H is continuous across the boundary

9. Give the force on a current element

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The force on a current element Idl is given by
 $dF = I \times B dl$

$$\sin = BI dl$$

10. Define magnetic dipole

A small bar magnet with pole strength Q_m and length l may be treated as magnetic dipole whose magnetic moment is $Q_m l$

11. Define hysteresis

The phenomenon which causes magnetic flux density to lag behind magnetic field intensity so that the magnetization curve for increasing and decreasing applied fields is not the same, is called hysteresis

-

It is used to find the field due to current J

13. Define magnetic dipole moment

A bar magnet of pole strength Q_m and length L constitutes a magnetic dipole of magnetic dipole moment $Q_m L$

$$\text{Magnetic dipole moment } m = Q_m L (\text{A.m}^2)$$

14. Define magnetic moment

Magnetic moment is defined as the maximum torque per magnetic induction (flux density)

$$m = IA$$

15. What is the relation between magnetic flux density and field intensity?

$$B = \mu_0 \mu_r H$$

16. What is field due to toroid and solenoid?

$$\text{Toroid } H = NI/2\pi r$$

$$\text{Solenoid } H = NI/l$$

17. Write down the expression for magnetic field at the centre of the circular coil $H = I/2a$

18. Define magnetic vector potential

It is defined as that quantity whose curl gives the magnetic flux density

Unit-3

1. What are the significant physical differences between Poisson's and Laplace's equations.

Poisson's and Laplace's equations are useful for determining the electrostatic potential V in regions whose boundaries are known. When the region of interest contains charges Poisson's equation can be used to find the potential.

$$\nabla \cdot \nabla V = \nabla^2 V = -\frac{\rho}{\epsilon_0}$$

When the region is free from charge Laplace equation is used to find the potential.

$$\nabla^2 V = 0$$

2. Brief about boundary conditions for electric fields.

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i) The tangential component of the electric field is continuous at the surface. $E_{t1} = E_{t2}$

ii) The normal component of the electric flux density is continuous if there is no surface charge density.

$$D_{n1} = D_{n2}$$

3. Define dielectric strength.

The dielectric strength of a dielectric is defined as the maximum value of electric field that can be applied to the dielectric without its electric breakdown.

4. What do you mean by magnetization?

Magnetization can be defined according to the following equation:

$$\mathbf{M} = \frac{N}{V} \mathbf{m} = n \mathbf{m}$$

Here, M represents magnetization; m is the vector that defines the magnetic moment; V represents volume; and N is the number of magnetic moments in the sample. The quantity N/V is usually written as n , the number density of magnetic moments.

5. What is meant by dielectric breakdown?

When the electric field in a dielectric is sufficiently large, it begins to pull electrons completely out of the molecules & the dielectric becomes conducting.

6. What is a homogeneous material?

Homogeneous material is one for which the quantities permeability, permittivity are constant throughout the medium.

7. Name the magnetic materials. Diamagnetic
Paramagnetic
Ferromagnetic
Ferrimagnetic.

8. Derive the expression for capacitance between two parallel plates.

$$C = \epsilon A / d$$

9. What is Lorentz force?

Lorentz force is the force experienced by the test charge. It is maximum if the direction of movement of charge is perpendicular to the orientation of field lines.

$$F = Q_1 Q_2 / 4\pi\epsilon r^2 \text{ ar}$$

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Unit-IV

1. What is skin effect?

In good conductors, the wave attenuates very rapidly & the fields are confined to the region near the surface of the conductor. This phenomenon is called the skin effect.

2. Define Brewster angle.

Brewster's angle is the angle at which the reflected light is linearly polarized normal to the plane incidence. At the end of the plasma tube, light can leave through a particular angle (Brewster's angle) and essentially be highly polarized. Maximum polarization occurs when the angle between reflected and transmitted light is 90°

thus $\theta_r + \theta_t = 90^\circ$; since $\sin(90-x) = \cos x$

Snell's provides $(\sin \theta_t / \cos \theta_r) = n_2/n_1$; θ_r is Brewster's angle

$$\theta_r = \tan^{-1} (n_2/n_1)$$

3. Define skin depth

It is defined as that depth in which the wave has been attenuated to $1/e$ or approximately 37% of its original value.

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$$\Delta = 1/\alpha = 2 / j\omega\sigma$$

4. Define Brewster angle.

Brewster's angle is the angle at which the reflected light is linearly polarized normal to the plane incidence. At the end of the plasma tube, light can leave through a particular angle (Brewster's angle) and essentially be highly polarized. Maximum polarization occurs when the angle between reflected and transmitted light is 90°

thus $\theta_r + \theta_t = 90^\circ$; since $\sin(90-x) = \cos x$

Snell's provides $(\sin \theta_i / \cos \theta_i) = n_2/n_1, \theta_r$ is Brewster's angle

5. What is skin effect?

In good conductors, the wave attenuates very rapidly & the fields are confined to the region near the surface of the conductor. This phenomenon is called the skin effect.

6. What is meant by circular polarization?

If x and y component of electric field E_x & E_y have equal amplitude and 90° phase difference, the locus of the resultant electric field E is a circle & the wave is a circularly polarized.

7. Brief about intrinsic impedance.

It is the ratio of electric field to magnetic field or It is the ratio of square root of Permeability to permittivity of medium.

8. What is meant by linear polarization?

If x and y component of electric field E_x & E_y are present and in phase, the locus of the resultant electric field E has a direction at an angle of $\tan^{-1}(E_y/E_x)$ & the wave is a linearly polarized.

9. Define wave

If a physical phenomenon that occurs at one place at a given time is reproduced at other places at later times, the time delay being proportional to the space separation from the first location, then the group of phenomena constitutes a wave

10. Mention the properties of uniform plane wave

The properties of uniform plane wave are as

follows

1. At every point in space, the electric field E and magnetic field H are perpendicular to each other and to the direction of the travel
2. The fields vary harmonically with time and at the same frequency, every where in space
3. Each field has the same direction, magnitude and phase at every point in any plane perpendicular to the direction of wave travel

11. Define intrinsic impedance or characteristic impedance

It is the ratio of electric field to magnetic field. Or it is the ratio of square root of permeability to permittivity of the medium

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UNIT I - STATIC ELECTRIC FIELD**Part A - Two Marks****1. Define scalar field?**

A field is a system in which a particular physical function has a value at each and every point in that region. The distribution of a scalar quantity with a defined position in a space is called scalar field.

Ex: Temperature of atmosphere.

2. Define Vector field?

If a quantity which is specified in a region to define a field is a vector then the corresponding field is called vector field.

3. Define scaling of a vector?

This is nothing but, multiplication of a scalar with a vector. Such a multiplication changes the magnitude of a vector but not the direction.

4. What are co-planar vector?

The vectors which lie in the same plane are called co-planar vectors.

5. Define base vectors?

The base vectors are the unit vectors which are strictly oriented along the directions of the coordinate axes of the given coordinate system.

6. What is a position vector?

Consider a point $p(x, y, z)$ are Cartesian coordinate system. Then the position vector of point p is represented by the distance of point p from the origin directed from origin to point. This is also called as radius vector.

7. Define Divergence.

Divergence is defined as the net outward flow of the flux per unit volume over a closed incremental surface.

8. State Divergence Theorem.

The integral of the normal component of any vector field over a closed surface is equal to the integral of the divergence of this vector field throughout the volume enclosed that closed surface.

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9. What is physical significance of curl of a vector field?

Curl gives rate of rotation. Curl F gives work done per unit area.

10. What is physical significance of divergence?

Divergence of current density gives net outflow of current per unit volume. Divergence of flux density gives net outflow per unit volume. In general, divergence of any field density gives net outflow of that field per unit volume.

11. State the conditions for a field to be a) solenoidal b) irrotational.

a) Divergence of the field has to be zero.

b) Curl of the field has to be zero.

12. Define scalar and vector quantity?

The scalar is a quantity whose value may be represented by a single real number which may be positive or negative. e.g, temperature, mass, volume, density.

A quantity which has both a magnitude and a specified direction in space is called a vector. e.g. force, velocity, displacement, acceleration.

13. What is a unit vector? What is its function while representing a vector?

A unit vector has a function to indicate the direction. Its magnitude is always unity, irrespective of the direction which it indicates and the coordinate system under consideration.

14. Name 3 coordinate systems used in electromagnetic engineering?

1) Cartesian or rectangular coordinate system.

2) Cylindrical coordinate system.

3) Spherical coordinate system.

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15. How to represent a point in a Cartesian system?

A point in rectangular coordinate system is located by three coordinates namely x, y and z coordinates. The point can be reached by moving from origin, the distance x in x direction then the distance y in y direction and finally z in z direction.

16. What is separation of vector?

The distance vector is also called as separation vector. Distance vector is nothing but the length of the vector.

17. State Distance formula?

Distance formula give the distance between the two points representing tips of the vector.

18. Show how a point p represented in a spherical coordinate system.

The point p can be defined as the intersection of three surfaces in spherical coordinate system.

r - Constant which is a sphere with centre as origin?

θ - Constant which is a right circular cone with apex as origin and axis as z axis?

Φ - Constant is a plane perpendicular to xy plane.

19. State the relationship between Cartesian and spherical system? $x = r \sin \theta \cos \Phi$

$$y = r \sin \theta \sin \Phi$$

$$z = r \cos \theta$$

Now r can be expressed as

$$x^2 + y^2 + z^2 = r^2 \sin^2 \theta \cos^2 \Phi + r^2 \sin^2 \theta \sin^2 \Phi + r^2 \cos^2 \theta$$

$$= r^2 \sin^2 \theta [\sin^2 \Phi + \cos^2 \Phi] + r^2 \cos^2 \theta$$

$$= r^2 [\sin^2 \theta + \cos^2 \theta]$$

$$= r^2$$

20. What are the types of integral related to electromagnetic theory?

1. Line integral
2. Surface integral
3. Volume integral

21. Give the types of charge distribution.

1. Line charge
2. Point charge
3. Surface charge
4. Volume charge.

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SRI VIDYA COLLEGE OF ENGINEERING & TECHNOLOGY Course Material(Question Bank)

Part B - Sixteen Marks

1. (a) State and prove divergence theorem.
 - Definition
 - Write Proof
2. Explain three co-ordinate systems.
 - Explain the details of Cartesian, Circular & Spherical Coordinate Systems
3. State and prove Gauss law and explain applications of Gauss law.
 - State Gauss law
 - Give the Proof
 - Write the application Field due to point charge, Field due to infinite line charge, Field due to infinite sheet of charge & Field due to coaxial cable.
4. Define the potential difference and absolute potential. Give the relation between potential and field intensity.
 - Derive the derivation for absolute potential
 - Potential due to N Charges
 - Derive relation between potential and electric field
5. Derive an expression for potential due to infinite uniformly charged line and also derive potential due to electric dipole.
 - Derive the expression for potential due to infinite uniformly charged line
 - Derive the electric dipole
6. State and prove stoke's theorem.
 - Give the definition of stokes theorem
 - Write the proof
7. State and explain coulomb's law.
 - State the coulombs law
 - Derive the expression & give the application

UNIT II – CONDUCTORS AND DIELECTRICS**Part A - Two Marks****1. Define point charge.**

A point charge means that electric charge which is separated on a surface or space whose geometrical dimensions are very small compared to other dimensions, in which the effect of electric field to be studied.

2. Define one coulomb.

One coulomb of charge is defined as the charge possessed by $(1/1.602 \times 10^{-19})$ i.e 6×10^{18} number of electrons.

3. What are the various types of charge distribution? Give an example for each.

1. Point charge - Ex. Positive charge
2. Line charge - Ex. A sharp beam in a cathode ray tube.
3. Surface charge - Ex. The plate of a charged parallel plate capacitor.
4. Volume charge - Ex. The charged cloud.

4. State the assumptions made while defining a Coulomb's law.

- 1) The two charges are stationary.
- 2) The two charges are point charge.

5. What is an equipotential surface?

An equipotential surface is an imaginary surface in an electric field of a given charge distribution, in which all points on the surface are at the same electric potential.

6. What is an electric flux?

The total number of lines of force in any particular electric field is called electric flux. It is represented by the symbol ψ . Similar to the charge, unit of electric flux is also Coulomb.

7. Define electric flux density.

The net flux passing normal through the unit surface area is called electric flux density. It is denoted as D . It has a specified direction which is normal to the surface area under consideration hence it is a vector field.

8. State Gauss's Law.

The electric flux passing through any closed surface is equal to the total charge enclosed by that surface.

9. State the application of Gauss's law.

- 1) The Gauss's law can be used to find E and D for symmetrical charge distributions.
- 2) It is used to find the charge enclosed or the flux passing through the closed surface.

10. What is an equipotential surface?

An equipotential surface is an imaginary surface in an electric field of a given charge distribution, in which all points on the surface are at the same electric potential.

11. Define the unit of Potential difference.

The unit of potential difference is Volt. One Volt potential difference is one Joule of work done in moving unit charge from one point to other in the field.

12. Define potential difference.

The work done per unit charge in moving unit charge from B to A in the field E is called potential difference between the points B to A.

13. Define relaxation time.

The relaxation time τ is defined as the time required by the charge density to decay to 36.8% of its initial value.

14. What is Potential Gradient?

The rate of change of potential with respect to the distance is called potential gradient.

15. What is Gaussian surface? What are the conditions to be satisfied in special Gaussian surface?

The surface over which is the Gauss's law is applied is called Gaussian surface. Obviously such a surface is a closed surface and it has to satisfy the following conditions.

- 1) The surface may be irregular but should be sufficiently large so as to enclose the entire charge.
- 2) The surface must be closed.
- 3) At each point of the surface D is either normal or tangential to the surface.
- 4) The electric flux density D is constant over the surface at which D is normal.

16. What is Gradient of V?

The maximum value of rate of change of potential with distance dv/dL is called gradient of V

17. Define Absolute potential.

The work done in moving a unit charge from infinity (or from reference point at which potential is zero) to the point under the consideration against E is called absolute potential of that point.

18. What is Polarization?

The applied field E shifts the charges inside the dielectric to induce the electric dipoles. This process is called Polarization.

19. What is Polarization of Dielectrics?

Polarization of dielectric means, when an electron cloud has a centre separated from the nucleus. This forms an electric dipole. The dipole gets aligned with the applied field.

20. What is method of images?

The replacement of the actual problem with boundaries by an enlarged region or with image charges but no boundaries is called the method of images.

21. When is method of images used?

Method of images is used in solving problems of one or more point charges in the presence of boundary surfaces.

Part B - Sixteen Marks

1. Derive the boundary conditions of the normal and tangential components of electric field at the interface of two media with different dielectrics.

- Explain details of free space & dielectric boundary conditions
- Derive the expression

2. Derive an expression for the capacitance of a parallel plate capacitor having two dielectric media.

- Define capacitor
- Explain parallel plate capacitor & derive the expression

3. Prove Laplace's and Poisson's equations.

- Write the derivation for poisson's and laplace equation
- Application of poisson's and laplace equation

4. Explain current density & Continuity equation.

- Define current density & write relationship between current and current density
- Derive the continuity equation

5. Write short notes on Images.

- Give the details of images
- Derive the experssion

UNIT III – STATIC MAGNETIC FIELDS**Part A - Two Marks**

1. What is Magnetic Field?

The region around a magnet within which influence of the magnet can be experienced is called Magnetic Field.

2. What are Magnetic Lines of Force?

The existence of Magnetic Field can be experienced with the help of compass field. Such a field is represented by imaginary lines around the magnet which are called Magnetic Lines of Force.

3. State Stoke Theorem.

The line integral of F around a closed path L is equal to the integral of curl of F over the open surface S enclosed by the closed path L .

4. Define scalar magnetic Potential.

The scalar magnetic potential V_m can be defined for source free region where J i.e. current density is zero.

5. What is the fundamental difference between static electric and magnetic field lines?

There is a fundamental difference between static electric and magnetic field lines. The tubes of electric flux originate and terminate on charges, whereas magnetic flux tubes are continuous.

6. State Kirchoff's Flux law.

It states that the total magnetic flux arriving at any junction in a magnetic circuit is equal to the magnetic flux leaving that junction. Using this law, parallel magnetic circuits can be easily analyzed.

7. State Kirchoff's MMF law.

Kirchoff's MMF law states that the resultant MMF around a closed magnetic circuit is equal to the algebraic sum of products of flux and reluctance of each part of the closed circuit.

8. What is Magnetization?

The field produced due to the movement of bound charges is called Magnetization represented by M .

9. State Biot Savart Law.

The Biot Savart law states that, the magnetic field intensity dH produced at a point p due to a differential current element IdL is,

- 1) Proportional to the product of the current I and differential length dL .
- 2) The sine of the angle between the element and the line joining point p to the element and
- 3) Inversely proportional to the square of the distance R between point p and the element.

10. Describe what are the sources of electric field and magnetic field?

Stationary charges produce electric field that are constant in time, hence the term electrostatics. Moving charges produce magnetic fields hence the term magnetostatics.

11. Define Magnetic flux density.

The total magnetic lines of force i.e. magnetic flux crossing a unit area in a plane at right angles to the direction of flux is called magnetic flux density. Unit Wb/m^2 .

12. State Ampere's circuital law.

The line integral of magnetic field intensity H around a closed path is exactly equal to the direct current enclosed by that path.

13. Define Magnetic field Intensity.

Magnetic Field intensity at any point in the magnetic field is defined as the force experienced by a unit north pole of one Weber strength, when placed at that point. Unit: N/Wb .

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14. What is rotational and irrotational vector field?

If curl of a vector field exists then the field is called rotational. For irrotational vector field, the curl vanishes i.e. curl is zero.

15. Give the application of Stoke's theorem.

The Stoke's theorem is applicable for the open surface enclosed by the given closed path. Any volume is a closed surface and hence application of Stoke's theorem to a closed surface which enclosed certain volume produces zero answer.

Part B - Sixteen Marks

1. Derive the expression for magnetic field intensity and magnetic flux density due to finite and infinite line.

- Drive the magnetic field intensity and magnetic flux density due to finite and infinite line

2. Derive the expressions for magnetic field intensity and magnetic flux density due to circular coil.

- Drive the magnetic field intensity and magnetic flux density due to circular coil.

3.State Ampere's circuital law and explain any two applications of Ampere's circuital law.

- Define Ampere's circuital law
- Derive and proof the point from of ampere's circuital law
- Give some application of ampere's law

4.State and explain Biot-Savart's law.

- Define Biot-Savart's law
- Derive the expression

5. State and prove stoke's theorem

- Define stoke's theorem and derive the expression.

UNIT IV – MAGNETIC FORCES AND MATERIALS**Part A - Two Marks****1. Define Inductance.**

In general, inductance is also referred as self inductance as the flux produced by the current flowing through the coil links with the coil itself.

2. What is fringing effect?

If there is an air gap in between the path of the magnetic flux, it spreads and bulges out. This effect is called fringing effect.

3. Define self inductance.

Self inductance is defined as the rate of total magnetic flux linkage to the current through the coil.

4. What is Magnetostatics?

The study of steady magnetic field, existing in a given space, produced due to the flow of direct current through a conductor is called Magnetostatics

5. Define Right hand Thumb Rule and where it is used?

Right hand Thumb Rule states that, hold the current carrying conductor in the right hand such that the thumb pointing in the direction of current and parallel to the conductor, then curled fingers point in the direction of magnetic lines of flux around it. It is used to determine the direction of Magnetic field around a conductor carrying a direct current.

6. Define Right handed Screw Rule.

It states that, imagine a right handed screw to be along the conductor carrying current with its axis parallel to the conductor and tip pointing in the direction of the current flow. Then the direction of Magnetic field is given by the direction in which screw must be turned so as to advance in the direction of current flow.

7. Define Mutual inductance.

The mutual inductance between the two coils is defined as the ratio of flux linkage of one coil to the current in other coil.

8. Define Reluctance.

Reluctance R is defined as the ratio of the magneto motive force to the total flux.

9. What is Lorentz force equation?

Lorentz force equation relates mechanical force to the electrical force. It is given as the total force on a moving charge in the presence of both electric and magnetic fields.

10. Define Moment of force.

The Moment of a force or torque about a specified point is defined as the vector product of the moment arm R and the force F . It is measured in Nm.

11. Define Magnetic dipole moment.

The Magnetic dipole moment of a current loop is defined as the product of current through the loop and the area of the loop, directed normal to the current loop.

12. Give any two dissimilarities between electric and magnetic circuits.

- 1) In electric circuit the current actually flows i.e. there is a movement of electrons whereas in magnetic circuit, due to MMF, flux gets established and doesn't flow in the sense in which current flows.
- 2) The electric lines of flux are not closed. They start from positive charge and end on negative charge and the magnetic lines of flux are closed lines.

13. What is Curl?

The curl is a closed line integral per unit area as the area shrinks to a point. It gives the circulation per unit area i.e. circulation density of a vector about a point at which the area is going to shrink. The curl also gives the direction, which is along the axis through a point at which curl is defined.

14. What is permeability?

In magnetostatics, the B and H are related to each other through the property of the region in which current carrying conductor is placed. It is called permeability denoted as μ . It is the ability with which the current carrying conductor forces the magnetic flux through the region around it.

15. Distinguish between solenoid and toroid.

Solenoid is a cylindrically shaped coil consisting of a large number of closely spaced turns of insulated wire wound usually on a non magnetic frame. If a long slender solenoid is bent into the form of a ring and thereby closed on itself it becomes a toroid.

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Part B - Sixteen Marks

1. Derive the expression for inductance and solenoid
 - Explain and derive the equation for inductance of solenoid
2. Derive an expression for inductance of a toroid coil with N turns, Carrying current I and radius of toroid R.
 - Explain toroid and give the equation of toroid
3. Derive an expression for torque on a loop carrying current I.
 - Derive the torque equation
4. Derive an expression for inductance of transmission line.
 - Explain and derive the equation for transmission line
5. Discuss the magnetic boundary condition
 - Explain the magnetic boundary condition & derive the boundary condition equations
6. Derive an expression for force between two parallel wires carrying current in same direction.
 - Derive the expression force
 - Give the equation of two parallel wires carrying current in same direction.

UNIT V – TIME VARYING FIELDS AND MAXWELL'S EQUATIONS**Part A - Two Marks**

1. Define a wave.

If a physical phenomenon that occurs at one place at a given time is reproduced at other places at later times, the time delay being proportional to the space separation from the first location then the group of phenomena constitutes a wave.

2. Mention the properties of uniform plane wave.

- i) At every point in space, the electric field E and magnetic field H are perpendicular to each other.
- ii) The fields vary harmonically with time and at the same frequency everywhere in space.

3. Define intrinsic impedance or characteristic impedance.

It is the ratio of electric field to magnetic field or It is the ratio of square root of permeability to permittivity of medium.

4. Give the characteristic impedance of free space.

377ohms.

5. Define propagation constant.

Propagation constant is a complex number $\gamma = \alpha + j\beta$ where γ is propagation constant.

6. Define skin depth

It is defined as that depth in which the wave has been attenuated to $1/e$ or approximately 37% of its original value.

7. Define Poynting vector.

The pointing vector is defined as rate of flow of energy of a wave as it propagates. $P = E \times H$

8. State Poynting's Theorem.

The net power flowing out of a given volume is equal to the time rate of decrease of the energy stored within the volume conduction losses.

9. Give the difficulties in FDM.

FDM is difficult to apply for problems involving irregular boundaries and non-homogeneous material properties.

10. Explain the steps in finite element method.

- i) Discrimination of the solution region into elements.
- ii) Generation of equations for fields at each element.
- iii) Assembly of all elements.
- iv) Solution of the resulting system.

11. State Maxwell's fourth equation.

The net magnetic flux emerging through any closed surface is zero.

12. State Maxwell's Third equation

The total electric displacement through the surface enclosing a volume is equal to the total charge within the volume.

13. State the principle of superposition of fields.

The total electric field at a point is the algebraic sum of the individual electric field at that point.

14. Define loss tangent.

Loss tangent is the ratio of the magnitude of conduction current density to displacement current density of the medium.

15. Define reflection coefficient.

Reflection coefficient is defined as the ratio of the magnitude of the reflected field to that of the incident field.

16. Define transmission coefficient.

Transmission coefficient is defined as the ratio of the magnitude of the transmitted field to that of incident field.

17. What will happen when the wave is incident obliquely over dielectric – dielectric boundary?

When a plane wave is incident obliquely on the surface of a perfect dielectric part of the energy is transmitted and part of it is reflected. But in this case the transmitted wave will be refracted, that is the direction of propagation is altered.

18. What are uniform plane waves?

Electromagnetic waves which consist of electric and magnetic fields that are perpendicular to each other and to the direction of propagation and are uniform in plane perpendicular to the direction of propagation are known as uniform plane waves.

19. What is the significant feature of wave propagation in an imperfect dielectric?

The only significant feature of wave propagation in an imperfect dielectric compared to that in a perfect dielectric is the attenuation undergone by the wave.

20. What is the major drawback of finite difference method?

The major drawback of finite difference method is its inability to handle curved boundaries accurately.

21. Define power density.

The power density is defined as the ratio of power to unit area. Power density = power/unit area.

22. What is Normal Incidence?

When a uniform plane wave incidences normally to the boundary between the media then it is known as normal incidence.

23. What is called attenuation constant?

When a wave propagates in the medium, it gets attenuated. The amplitude of the signal reduces. This is represented by attenuation constant α . It is measured in Neper per meter (NP/m). But practically it is expressed in decibel (dB).

24. What is phase constant?

When a wave propagates, phase change also takes place. Such a phase change is expressed by a phase constant β . It is measured in radian per meter (rad/m).

25. How voltage maxima and minima are separated?

In general voltage minima are separated by one half wavelength. Also the voltage maxima are also separated by one half wave length.

Part B - Sixteen Marks

1. With necessary explanation, derive the Maxwell's equation in differential and integral forms.
 - Derive and explain Maxwell's equation I,II,III & IV
2. Write short notes on faradays law of electromagnetic induction.
 - Define faradays law
 - Derive the equation
3. State and explain the poynting theorem.
 - Define poynting theorem
 - Derive average power density, Integral and point form of poynting theorem & Instantaneous, average and complex poynting vector.
4. Explain transformer and motional electromotive force.
 - Write stationary conductor in time varying magnetic field & Moving conductor in satatic magnetic field.
5. Explain electromagnetic boundary condition & Wave equation.
 - Write and derive boundary condition
 - Write the wave equation for free space.

SVS COLLEGE OF ENGINEERING, COIMBATORE

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

EC6403 ELECTROMAGNETIC FIELDS

CLASS-SEM: II B.E. ECE - IVSEM

TWO MARKS

UNIT I- STATIC ELECTRIC FIELD

1. Define scalar field.

A field is a system in which a particular physical function has a value at each and every point in that region. The distribution of a scalar quantity with a defined position in a space is called scalar field.

Ex: Temperature of atmosphere and Electric potential.

2. Define Vector field.

If a quantity which is specified in a region to define a field is a vector then the corresponding field is called vector field.

Ex. Electric and magnetic Field Intensity.

3. What are co-planar vector?

The vectors which lie in the same plane are called co-planar vectors.

4. What is an identical vector?

Two vectors are said to be identical if their difference is zero. Thus A and B are identical if $A - B = 0$, i.e., $A = B$. Such two vectors are also called as equal vectors.

5. Define base vectors.

The base vectors are the unit vectors which are strictly oriented along the directions of the coordinate axes of the given coordinate system.

6. What is a position vector?

Consider a point $p(x, y, z)$ as Cartesian coordinate system, then the position vector of point p is represented by the distance of point p from the origin directed from origin to point. This is also called as radius vector.

7. Define scalar product of vectors.

The scalar of the two vectors A and B is denoted as $A \cdot B$ and defined as the product of the magnitude of A and magnitude of B and the cosine of angle between them.

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta_{AB}$$

where θ_{AB} - Angle between $\vec{A}$ & $\vec{B}$.

8. Define Divergence.

Divergence is defined as the net outward flow of the flux per unit volume over a closed incremental surface.

9. State Divergence Theorem.

The integral of the normal component of any vector field over a closed surface is equal to the integral of the divergence of this vector field throughout the volume enclosed that closed surface.

$$\int_V (\nabla \cdot \vec{A}) dv = \oint_S \vec{A} \cdot d\vec{S}$$

10. Define curl of a vector.

The maximum circulation of $\vec{F}$ per unit area as area tends to zero whose direction is normal to the surface is called curl of $\vec{F}$. $\text{Curl}(\vec{F}) = \nabla \times \vec{F} = \lim_{\Delta S \rightarrow 0} \left(\frac{\oint_L \vec{F} \cdot d\vec{r}}{\Delta S} \right) \hat{n}$.

11. State Stoke Theorem.

The line integral of $\vec{F}$ around a closed path L is equal to the integral of curl of $\vec{F}$ over the open surface S enclosed by the closed path L .

$$\oint_L \vec{F} \cdot d\vec{r} = \int_S (\nabla \times \vec{F}) \cdot d\vec{S}$$

12. What is physical significance of curl of a vector field?

Curl gives rate of rotation. $\text{Curl } \vec{F}$ gives work done per unit area. where $\vec{F}$ is vector field.

13. What is physical significance of divergence?

Divergence of current density gives net outflow of current per unit volume. Divergence of flux density gives net outflow of flux per unit volume. In general, divergence of any field density gives net outflow of that field per unit volume.

14. State the conditions for a field to be a) solenoidal b) irrotational.

- a) Divergence of the field has to be zero. $\nabla \cdot \vec{A} = 0$
 b) Curl of the field has to be zero. $(\nabla \times \vec{A}) = 0$ where $\vec{A}$ is vector field.

15. Name the three coordinate systems used in electromagnetic engineering.

- Cartesian or rectangular coordinate system.
- Cylindrical coordinate system.
- Spherical coordinate system.

16. State the relation between Cartesian and cylindrical coordinate system.

$$x = r \cos \phi; y = r \sin \phi; z = z; \quad \rho = \sqrt{x^2 + y^2}; \quad \phi = \tan^{-1}(y/x); \quad z = z$$

17. State the relationship between Cartesian and spherical system.

$$x = r \sin \theta \cos \Phi; y = r \sin \theta \sin \Phi; z = r \cos \theta$$

$$r = \sqrt{x^2 + y^2 + z^2}; \quad \theta = \tan^{-1}\left(\frac{\sqrt{x^2 + y^2}}{z}\right); \quad \phi = \tan^{-1}(y/x)$$

18. State dot product properties.

- It obeys commutative law. $A \cdot B = B \cdot A$
- It obeys distributive law. $A \cdot (B + C) = A \cdot B + A \cdot C$

3) If the dot product with itself is performed the result is square of the magnitude of that vector $A \cdot A = |A|^2$

4) Any unit vector dotted with itself is unity. $\hat{a}_x \cdot \hat{a}_x = \hat{a}_y \cdot \hat{a}_y = 1$

19. What is called as cross product?

Cross product is also called as vector product. It is defined as the product of the magnitude of A and magnitude of B and sine of the smallest angle between them.

$$\vec{A} \times \vec{B} = |\vec{A}| |\vec{B}| \sin \theta \hat{n} ; \quad \theta \rightarrow \text{Angle b/w } \vec{A} \text{ \& } \vec{B}$$

$\hat{n}$ - normal to the $\vec{A}$ \& $\vec{B}$.

20. What are the types of integral related to electromagnetic theory?

Line integral, Surface integral and Volume integral

21. Define point charge.

A point charge means that electric charge which is separated on a surface or space whose geometrical dimensions are very small compared to other dimensions, in which the effect of electric field to be studied.

22. State Coulomb's law.

The coulomb's law states that force between the two point charges Q_1 and Q_2 is

- i) Acts along the line joining the two point charges
- ii) Directly proportional to the product of the charges
- iii) Inversely proportional to the square of the distance between them.

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{R^2} \hat{a}_1 ; \quad \epsilon_0 - \text{Permittivity of free space.}$$

Q_1, Q_2 - Point charges.

23. What are the various types of charge distribution? Give an example for each.

- Point charge-Ex. Positive charge
- Line charge -Ex. A sharp beam in a cathode ray tube.
- Surface charge-Ex. The plate of a charged parallel plate capacitor.
- Volume charge-Ex. The charged cloud.

24. State the assumptions made while defining a Coulomb's law.

- The two charges are stationary.
- The two charges are point charge.

25. Define electric field intensity at a point.

The electric field intensity is defined as force exerted per unit charge.

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 R^2} \hat{a}_R$$

26. Define conservative field.

The sign indicates integral over a closed path. Such a field having property of $\oint \vec{E} \cdot d\vec{L} = 0$ associated with it is called conservative field or lamellar field.

27. What is an equipotential surface?

An equipotential surface is an imaginary surface in an electric field of a given charge distribution, in which all points on the surface are at the same electric potential.

28. What is an electric flux?

The total number of lines of force in any particular electric field is called electric flux. It is represented by the symbol Φ . Similar to the charge unit of electric flux is also Coulomb.

29. Define electric flux density.

The net flux passing normal through the unit surface area is called electric flux density. It is denoted as D . It has a specified direction which is normal to the surface area under consideration hence it is a vector field.

30. State the vector form of electric flux density.

The flux density $\vec{D}$ at the point p can be represented in the vector form as

$$\vec{D} = \frac{Q}{4\pi R^2} \hat{a}_R \quad \begin{array}{l} Q - \text{Point charge,} \\ R - \text{Distance between origin and point } p. \end{array}$$

31. State the relationship between D and E .

$$\vec{D} = \epsilon \vec{E} \quad ; \quad \begin{array}{l} \text{where } \epsilon - \text{permittivity of material in electric field,} \\ \vec{D} - \text{flux density ; } \vec{E} - \text{field intensity.} \end{array}$$

32. State Gauss's Law.

The electric flux passing through any closed surface is equal to the total charge enclosed by that surface.

$$\phi = Q_{enc} \quad \text{where } \phi - \text{electric flux ; } Q - \text{Total charge}$$

$$Q_{enc} = \oint_V \vec{D} \cdot d\vec{s} \quad ; \quad \phi = \int_V \rho_v dv$$

$$\nabla \cdot \vec{D} = \rho_v \quad ; \quad \text{where } \rho_v - \text{volume charge density.}$$

33. State the application of Gauss's law.

- 1) The Gauss's law can be used to find E and D for symmetrical charge distributions.
- 2) It is used to find the charge enclosed or the flux passing through the closed surface.

34. What is Potential Gradient?

The rate of change of potential with respect to the distance is called potential gradient. i.e., ∇V gradient of potential.

35. What is Gaussian surface? What are the conditions to be satisfied in special Gaussian surface?

The surface over which is the Gauss's law is applied is called Gaussian surface. Obviously such a surface is a closed surface and it has to satisfy the following conditions.

- 1) The surface may be irregular but should be sufficiently large so as to enclose the entire charge.
- 2) The surface must be closed.
- 3) At each point of the surface D is either normal or tangential to the surface.
- 4) The electric flux density D is constant over the surface at which D is normal.

36. Define Absolute potential.

The work done in moving a unit charge from infinity (or from reference point at which potential is zero) to the point under the consideration against E is called absolute potential of that point

37. What is the relationship between V and E ?

$$E = -\text{grad}V \quad ; \quad \vec{E} = -\nabla V \quad ; \quad V = -\int_L \vec{E} \cdot d\vec{r}$$

38. Define dipole and dipole moment.

Dipole or electric dipole is nothing but two equal and opposite point charges are separated by a very small distance.

The product of electric charge and distance is known as dipole moment. It is denoted by m where Q is the charge and L is the length.

$$M = QL \text{ coulomb meter}$$

39. Write the expression for the energy density in electrostatic field.

$$\frac{\Delta w}{\Delta v} = \frac{1}{2} \epsilon E^2 = \frac{1}{2} DE$$

ϵ - permittivity of electric field.

D - flux density

E - field intensity.

$\frac{\Delta w}{\Delta v}$ - energy stored per volume = energy density.

UNIT II -CONDUCTORS & DIELECTRICS

1. Define relaxation time.

The relaxation time τ is defined as the time required by the charge density to decay to 36.8% of its initial value.

$\tau = \epsilon / \sigma$ where τ is relaxation time, ϵ is permittivity of the material and σ is conductivity of the material.

2. State Uniqueness Theorem.

The Uniqueness theorem can be stated as, The solution of Laplace's equation gives the field which is unique satisfying the same boundary conditions, in a given region.

3. State the applications of Poisson's equation and Laplace's equation.

- 1) To obtain potential distribution over the region.
- 2) To obtain E in the region.
- 3) To check whether given region is free of charge or not.
- 4) To obtain the charge induced on the surface of the region.

4. Define current density (J).

The current density is defined as the current passing through the unit surface area, when the surface is held normal to the direction of the current. The current density is measured in A/m^2 .

5. Define a current and its unit Ampere.

The current is defined as the rate of flow of charge and is measured as Ampere. A current of 1 Ampere is said to be flowing across the surface when the charge of 1 coulomb is passing across the surface in 1 second.

6. What is convection current?

In dielectrics, there can be flow of charges, under the influence of electric field intensity. Such a current is called convection current.

7. State the principle of conservation of charge.

The principle of conservation of charge is, the charges can neither be created nor be destroyed.

8. Define dielectric strength.

The minimum value of the applied electric field at which the dielectric breaks down is called dielectric strength of dielectric.

9. What is Polarization?

The applied field E shifts the charges inside the dielectric to induce the electric dipoles. This process is called Polarization.

10. State the point form of Ohm's law.

The relationship between $\vec{J}$ and $\vec{E}$ can also be expressed in terms of conductivity of the material. Thus for metallic conductor, $\vec{J} = \sigma \vec{E}$; where J is current density, σ is conductivity of material and E is Electric field intensity

The field strength within a conductor is proportional to the current density. The equation is called point form of Ohm's law.

11. What is Boundary conditions means?

The conditions existing at the boundary of the two media when field passes from one medium to other are called boundary conditions.

12. What are dielectrics?

Dielectrics are materials that may not conduct electricity through it but on applying electric field induced charges are produced on its faces. The valence electron in atoms of a dielectric are tightly bound to their nucleus.

13. What is a capacitor?

A capacitor is an electrical device composed of two conductors which are separated through a dielectric medium and which can store equal and opposite charges, independent of whether other conductors in the system are charged or not.

14. What are the factors does the capacitance depends on?

- The permittivity of the dielectric used.
- The area of cross section of the plates
- The distance of separation of the plates

15. What is meant by displacement current?

Displacement current is nothing but the current flowing through the capacitor.

16. What are the two situations of the boundary conditions based on nature of the media?

Boundary between conductor and free space.

Boundary between two dielectrics with different properties.

17. State Poisson's equation.

The Poisson's equation is stated as, $\nabla^2 v = \frac{\rho_v}{\epsilon}$

where $\nabla^2 v = \nabla \cdot (\nabla v)$ - Divergence of Gradient of potential v
 ρ_v - Volume charge density; ϵ - Permittivity of electric field.

18. State Laplace's equation.

The Laplace's equation is stated as,

$\nabla^2 v = 0$ where ∇^2 - Laplacian operator.

$\nabla^2 v = \nabla \cdot (\nabla v) \Rightarrow$ Divergence of gradient of potential v

19. Write the expression for continuity equation and explain its significance.

$\nabla \cdot \vec{J} = -\frac{\partial \rho_v}{\partial t}$; $\vec{J}$ - Current density
 ρ_v - volume charge density.

Since the charge is conserved, the outward flux of J must therefore be equal to the rate of loss of charge within the volume.

20. Write the capacitance expression for a parallel plate capacitor.

$C = \frac{\epsilon A}{d}$ Farads ; C - Capacitance b/w parallel plates.
 ϵ - Permittivity of electric field ; A - Area of the plate
 d - distance between the two parallel plates.

21. Write the capacitance expression for a coaxial cable.

$C = \frac{2\pi\epsilon L}{\ln(b/a)}$ Farad L - length of the cable.
 a & b are the radius of the inner and outer cable respectively.
 ϵ - permittivity of electric field.

22. Write the capacitance expression for two concentric spheres.

$C = \frac{4\pi\epsilon}{1/a - 1/b}$; a & b are radius of inner and outer sphere respectively.
 ϵ - permittivity of electric field.

23. Write the expression for conduction current and current density.

$I_c = \sigma \vec{E} \cdot A$; $J_c = \frac{I_c}{A} = \sigma \vec{E}$ $\vec{E}$ - field intensity.
 I_c - Conduction Current ; J_c - current density.
 A - Area of the conductor ; σ - conductivity.

24. Write the expression for displacement current and current density.

$I_d = A \frac{\partial \vec{D}}{\partial t}$; $J_d = \frac{I_d}{A} = \frac{\partial \vec{D}}{\partial t}$ $\vec{D}$ - flux density.
 I_d - displacement current ; J_d - displacement current density

25. State the boundary conditions at the interface between two perfect dielectrics.

- The tangential component of the electric field is continuous ie, $E_{t1} = E_{t2}$
- The normal component of the flux density is continuous ie, $D_{n1} = D_{n2}$

26. State the boundary conditions at the interface between two perfect conductors.

- The tangential component of the ^{flux density} is continuous ie, $D_{t1} = D_{t2}$
- The normal component of the electric field is continuous ie, $E_{n1} = E_{n2}$.

27. What are the significant physical difference between Poisson's & Laplace equation?

Poisson's and Laplace equation are useful for determining the electrostatic potential V in regions at whose boundaries are known.

When the region contains charges in a known distribution, then the Poisson's equation can be used to determine the potential function.

When the region is free from charge, Poisson's equation becomes Laplace equation.

28. Define method of images.

A point charge Q at a distance from grounded conducting plane, the positive charge induces another charge with opposite direction under the conducting plane. This induced charge is referred as electrical image. This is called method of images.

29. Write the expression for resistance of the conductor of uniform and non uniform cross section.

Uniform Cross section $R = \frac{\rho_c l}{A}$; l - length of the conductor
 ρ_c - resistivity. A - cross section area.
 Non-Uniform Cross section $R = \frac{\int \vec{E} \cdot d\vec{l}}{\int \vec{J} \cdot d\vec{S}}$; $\vec{E}$ - field intensity.
 $\vec{J}$ - current density.

30. Give the boundary condition for conductor - conductor interface medium.

The Boundary Conditions are $\vec{D}_t = \epsilon_0 \epsilon_r \vec{E}_t = 0$;
 $\vec{D}_n = \epsilon_0 \vec{E}_n = \rho_s$ where ρ_s - surface charge density.

31. Give the boundary condition for conductor - free space interface medium.

The Boundary Conditions are $\vec{D}_t = \epsilon_0 \vec{E}_t = 0$
 $\vec{D}_n = \epsilon_0 \vec{E}_n = \rho_s$ Here $\epsilon_r = 1$ - dielectric constant in free space.

32. Define linear, isotropic and homogeneous medium.

- The medium is said to be linear if D vary linearly with E .
- The medium is said to be isotropic, if D and E are in same direction.
- The medium is said to be homogeneous, if ϵ is independent of space coordinates or uniform over the entire region.

33. Define dielectric constant.

The ratio of permittivity of the dielectric to that of free space

$$\text{Dielectric Constant } \epsilon_r = \frac{\epsilon}{\epsilon_0} = \frac{\text{permittivity of dielectric}}{\text{permittivity of free space.}}$$

UNIT III – STATIC MAGNETIC FIELDS

1. Define Magnetic flux density.

The total magnetic lines of force i.e. magnetic flux crossing a unit area in a plane at right angles to the direction of flux is called magnetic flux density. It is denoted as B , Unit is Wb/m^2 . $\vec{B} = \frac{\phi}{A}$ where ϕ - Magnetic flux

$\vec{B}$ - flux density A - Area.

2. Define Magnetic field Intensity.

Magnetic Field intensity at any point in the magnetic field is defined as the force experienced by a unit north pole of one Weber strength, when placed at that point. Unit is N/Wb (or) A/m . It is denoted as H .

3. Give the application of Stoke's theorem.

The Stoke's theorem is applicable for the open surface enclosed by the given closed path. Any volume is a closed surface and hence application of Stoke's theorem to a closed surface which enclosed certain volume produces zero answer.

4. State Ampere's circuital law.

The line integral of magnetic field intensity $\vec{H}$ around a closed path is exactly equal to the direct current enclosed by that path.

$$\oint_L \vec{H} \cdot d\vec{L} = I_{enc}$$

$\vec{H}$ - Magnetic field intensity

5. What is rotational and irrotational vector field?

If curl of a vector field exists then the field is called rotational. For irrotational vector field, the curl vanishes i.e. curl is zero.

6. State Biot Savart Law.

The magnetic field intensity $d\vec{H}$ produced at a point p due to a differential current element $I d\vec{L}$ is proportional to the product of the current I and differential length dL , the sine of the angle between the element and the line joining point p to the element and inversely proportional to the square of the distance R between point p and the element.

$$d\vec{H} = \frac{I dL \sin \alpha}{4\pi R^2}$$

7. Give the relation between Magnetic flux and Flux density.

The relation between Magnetic flux and flux density is obtained through the property of medium and permeability μ . $\vec{B} = \mu \vec{H}$

8. State Law of conservation of Magnetic Flux.

It states that, the integral, $\vec{B} \cdot d\vec{S}$ is over a closed surface is always zero.

$$\int \vec{B} \cdot d\vec{S} = 0$$

This is also called Gauss's law in integral form for magnetic fields.

9. Give Gauss's law in differential form for magnetic fields.

The divergence of magnetic flux density is always zero.

$$\nabla \cdot \vec{B} = 0$$

10. Define scalar magnetic Potential.

The scalar magnetic potential V_m can be defined for source free region where J i.e. current density is zero.

$$\vec{H} = -\nabla V_m \quad H - \text{Magnetic field Intensity}$$

V_m - Magnetic potential.

11. Define current density.

Current density is defined as the current per unit area. $J = I/A$. A/m^2

J - Current density ; I - current ; A - Area.

12. Define magnetic vector potential.

It is defined as that quantity whose curl gives the magnetic flux density.

$$\vec{B} = \nabla \times \vec{A}$$

Where A is the magnetic vector potential.

$$A = \frac{\mu}{4\pi} \iiint \frac{J}{r} dr \text{ wb/m}$$

μ - Permeability of Magnetic field.

J - Current density.

13. Give point or differential form of ampere's circuit law.

Curl of the magnetic field intensity is current density. $\nabla \times \vec{H} = \vec{J}$

$\vec{H}$ - Magnetic field intensity ;

$\vec{J}$ - Current density.

14. Write the applications of ampere's circuit law.

The applications of ampere's circuit law are to find H for infinitely long straight conductor, co-axial cable and infinite sheet of current.

15. State the expressions of H due to straight conductor carrying a current I of finite length and infinite length.

$$\text{Finite length, } \vec{H} = \frac{I}{4\pi\rho} (\cos \alpha_2 - \cos \alpha_1) \hat{a}_z$$

$$\text{Infinite length, } \vec{H} = \frac{I}{2\pi\rho} \hat{a}_z$$

ρ - perpendicular distance ; I - current

16. Give the expression for magnetic field intensity at any point on the axis of the circular loop.

$$\vec{H} = \frac{I\rho^2}{2(h^2 + \rho^2)^{3/2}} \hat{a}_z$$

Here ρ - radius of the circular loop

h - distance b/w point P & origin on the z axis.

17. Give the expressions relating B and H with the current density J.

$$\vec{B} = \frac{\mu_0}{4\pi} \iiint \frac{J \times \vec{a}_r}{r^2} dv; \text{curl} \vec{B} = \mu_0 \vec{J}; \text{curl} \vec{H} = \vec{J}$$

$\vec{H}$ - Magnetic field Intensity

$\vec{B}$ - Magnetic flux Density

$\vec{J}$ - Current density.

μ_0 - permeability in free space.

UNIT IV MAGNETIC FORCES AND MATERIALS

1. Classify the materials depending on their magnetic properties.

The materials are classified based on their magnetic properties as follows, Diamagnetic, Paramagnetic and Ferromagnetic.

2. How will you identify the type of a magnetic material?

The magnetic materials can be identified using the relative permeability (μ_r).

If $\mu_r \leq 1$, the materials are diamagnetic

If $\mu_r \geq 1$, the materials are paramagnetic

If $\mu_r \gg 1$, the materials are ferromagnetic

3. What is a magnetic dipole?

Two equal and opposite magnetic poles separated by a small distance in the form of a bar magnet is called magnetic dipole.

4. What is Magnetization (M)?

The alignment of the magnetic dipoles along the direction, in which magnetic field H is applied is called magnetization. It is also equivalent to the dipole moment per unit volume. Q_m - Magnetic charge (or) pole strength, l - length.

$$M = \frac{Q_m l}{Al} ; \quad \text{Therefore, } M = \frac{Q_m}{A} \hat{a} \text{ A/m} \quad A - \text{Area}$$

Where, Q_m is the pole strength, A is the area of cross-section of the bar magnet and 'l' is the length of the bar magnet.

5. Define magnetic susceptibility.

Magnetic susceptibility is defined as the ratio of magnetization to magnetic field

$$\text{intensity, } X_m = \frac{M}{H} \quad \begin{array}{l} M - \text{Magnetization} \\ H - \text{Magnetic field Intensity} \end{array}$$

6. Give relation between relative permeability and magnetic susceptibility.

The relative permeability and magnetic susceptibility are related as,

$$\mu_r = 1 + X_m \quad \begin{array}{l} X_m \rightarrow \text{Magnetic susceptibility} \\ \mu_r - \text{Relative permeability} \end{array}$$

7. What is inductance?

The inductance is defined as the rate of total magnetic flux linkage to the current through the coil and is generally denoted by the symbol L. The device which possesses this property is called inductor.

$$L = N \frac{d\phi}{di} \text{ Henry} \quad \begin{array}{l} N - \text{no. of turns.} \\ \phi - \text{Magnetic flux,} \\ L - \text{Inductance.} \end{array}$$

13. Write Lorentz-Force equation and mention its applications.

$$\vec{F} = Q(\vec{E} + (\vec{V} \times \vec{B})) \quad \vec{F} - \text{Force}$$

It is used to determine a motion of charged particle in a combined electric and magnetic field.

14. Define magnetic moment(m).

Magnetic moment is defined as the maximum torque per magnetic flux density.

$$m = T/B; \quad T - \text{Torque}; \quad B - \text{flux density}$$

15. Define Mutual Inductance.

Mutual inductance between two coils is defined as the ratio of flux linkage of one coil to current in the other coil. $M_{12} = \frac{N_2 \Phi_{12}}{I_1}$; N_2 - No. of turn in coil 2

$$\Phi_{12} - \text{flux between coil 1 \& 2.}$$

16. Write the expression for torque in vector form.

The torque on the loop is vector product of moment arm and force.

$$\vec{T} = \vec{m} \times \vec{B} = B I A \sin \theta.$$

17. Write the force equation on a current element.

The force on a current element Idl is given by $d\vec{F} = I d\vec{l} \times \vec{B}$

$$\vec{B} - \text{Magnetic flux density.}$$

18. Define Co-efficient of coupling.

The fraction of total flux produced by one coil linking the second coil is called co efficient of coupling (K).

$$K = \frac{\Phi_{12}}{\Phi_1} = \frac{\Phi_{21}}{\Phi_2}$$

19. How do mutual inductance between two coil is related to their self inductance?

$$M = K \sqrt{L_1 L_2}; \quad M - \text{Mutual Inductance.}$$

L_1, L_2 are Inductance of coil 1 & 2 respectively.

20. What will be effective inductance, if two inductors are connected in a series and parallel?

$$\text{For series}; \quad L = L_1 + L_2 \pm 2M$$

$$\text{For parallel}; \quad L = \frac{L_1 L_2 - M^2}{L_1 + L_2 \pm 2M}.$$

'+' indicates for Aiding.

'-' indicates for Opposing.

UNIT V TIME VARYING FIELDS & MAXWELL'S EQUATION

1. State Faraday's law.

Faraday's law states that, the total emf induced in a closed circuit is equal to the time rate of decrease of the total magnetic flux linking the circuit.

$$e = -\frac{d\phi}{dt} \text{ Volts}$$

2. Give the Maxwell's equation – I in both integral form and point form.

Maxwell's equation – I is derived from the Ampere's circuital law which states that the line integral of magnetic field intensity H on any closed path is equal to the current enclosed by that path.

$$\oint \vec{H} \cdot d\vec{l} = I$$

Maxwell's equation – I in integral form is $\oint H \cdot dl = \iint (\sigma E + \epsilon \frac{\partial E}{\partial t}) \cdot \vec{ds}$

Maxwell's equation – I in point form is $\nabla \times \vec{H} = \sigma \vec{E} + \epsilon \frac{\partial \vec{E}}{\partial t}$

The magneto motive force around a closed path is equal to the sum of the conduction current and displacement current enclosed by the path.

3. Give the Maxwell's equation – II in both integral form and point form.

Maxwell's equation – II is derived from Faraday's law which states that the emf induced in a circuit is equal to the rate of decrease of the magnetic flux linkage in the circuit.

$$e = -\frac{d\phi}{dt}$$

Maxwell's equation - II in integral form is $\oint \vec{E} \cdot d\vec{l} = -\mu \iint \frac{\partial \vec{H}}{\partial t} \cdot \vec{ds}$

Maxwell's equation – II in point form is $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

The electro motive force around a closed path is equal to the magnetic displacement (flux density) through that closed path.

4. Give the Maxwell's equation – III in both integral form and point form.

The Maxwell's equation – III is derived from electric Gauss's law which states that the electric flux through any closed surface is equal to the charge enclosed by the surface.

$$\psi = Q$$

Maxwell's equation – III in integral form is $\oint_S \vec{D} \cdot d\vec{s} = \iiint_V \rho dv$

Maxwell's equation – III in point form is $\nabla \cdot \vec{D} = \rho$

The total electric displacement through the surface enclosing a volume is equal to the total charge within the volume.

5. Give the Maxwell's equation – IV in both integral form and point form.

Maxwell's equation – IV is derived from magnetic Gauss's law which states that, the total magnetic flux through any closed surface is equal to zero.

$$\phi = 0$$

Maxwell's equation – IV in integral form is $\oint_S \vec{B} \cdot d\vec{s} = 0$

Maxwell's equation – IV in point form is $\nabla \cdot \vec{B} = 0$

The net magnetic flux emerging through any closed surface is zero.

6. Define Poynting vector.

The poynting vector is a vector that gives the relation between the rate of the energy transfer in an electromagnetic wave and the amplitudes of electric and magnetic field intensities of the electromagnetic wave.

$$\vec{P} = \vec{E} \times \vec{H}$$

7. State Poynting theorem.

The poynting theorem states that the vector product of electric field intensity and magnetic field intensity at any point is a measure of the rate of energy flow per unit area at that point.

$$\vec{P} = \vec{E} \times \vec{H}$$

The direction of flow P is perpendicular to E and H.

8. Distinguish between the conduction current and displacement current.

Conduction current I_c is flowing through a conductor having resistance R, when potential V is applied across the conductor.

$$I_c = \frac{V}{R}$$

Displacement current I_D is flowing through a capacitor when ac voltage is applied across the capacitor.

$$I_D = C \frac{dV}{dt}$$

9. What is the fundamental difference between static electric and magnetic field lines?

There is a fundamental difference between static electric and magnetic field lines. The tubes of electric flux originate and terminates on charges, whereas magnetic flux tubes are continuous.

10. What is called as intrinsic impedance?

The ratio of amplitudes of E and H of the waves in either direction is called intrinsic impedance of the material in which wave is travelling. It is denoted by η .

11. What is called wave velocity?

The velocity of propagation is called as wave velocity. It is denoted as v .

$$v = \frac{1}{\sqrt{\mu\epsilon}}$$

μ - permeability ; ϵ - permittivity,
 v - Wave velocity.

12. What are uniform plane waves?

Electromagnetic waves which consist of electric and magnetic fields that are perpendicular to each other and to the direction of propagation and are uniform in plane perpendicular to the direction of propagation are known as uniform plane waves.

13. Give the characteristic impedance of free space.

377ohms

14. Define a wave.

If a physical phenomenon that occurs at one place at a given time is reproduced at other places at later times, the time delay being proportional to the space separation from the first location then the group of phenomena constitutes a wave.

15. Mention the properties of uniform plane wave.

- i) At every point in space, the electric field E and magnetic field H are perpendicular to each other.
- ii) The fields vary harmonically with time and at the same frequency everywhere in space.

16. What is phase constant?

When a wave propagates, phase change also takes place. Such a phase change is expressed by a phase constant β . It is measured in radian per meter (rad/m).

Unit 1

Part A - Two Marks

1. Define scalar field?

A field is a system in which a particular physical function has a value at each and every point in that region. The distribution of a scalar quantity with a defined position in a space is called scalar field.

Ex: Temperature of atmosphere.

2. Define Vector field?

If a quantity which is specified in a region to define a field is a vector then the corresponding field is called vector field.

3. Define scaling of a vector?

This is nothing but, multiplication of a scalar with a vector. Such a multiplication changes the magnitude of a vector but not the direction.

4. What are co-planar vector?

The vectors which lie in the same plane are called co-planar vectors.

5. Define base vectors?

The base vectors are the unit vectors which are strictly oriented along the directions of the coordinate axes of the given coordinate system.

6. What is a position vector?

Consider a point $p(x, y, z)$ in Cartesian coordinate system. Then the position vector of point p is represented by the distance of point p from the origin directed from origin to point. This is also called as radius vector.

7. Define Divergence.

Divergence is defined as the net outward flow of the flux per unit volume over a closed incremental surface.

8. State Divergence Theorem.

The integral of the normal component of any vector field over a closed surface is equal to the integral of the divergence of this vector field throughout the volume enclosed that closed surface.

9. What is physical significance of curl of a vector field?

Curl gives rate of rotation. $\text{Curl } F$ gives work done per unit area.

10. What is physical significance of divergence?

Divergence of current density gives net outflow of current per unit volume. Divergence of flux density gives net outflow per unit volume. In general, divergence of any field density gives net outflow of that field per unit volume.

11. State the conditions for a field to be a) solenoidal b) irrotational.

a) Divergence of the field has to be zero. b) Curl of the field has to be zero.

12. Define scalar and vector quantity?

The scalar is a quantity whose value may be represented by a single real number which may be positive or negative. e.g, temperature, mass, volume, density. A quantity which has both a magnitude and a specified direction in space is called a vector. e.g, force, velocity, displacement, acceleration.

13. What is a unit vector? What is its function while representing a vector?

A unit vector has a function to indicate the direction. Its magnitude is always unity, irrespective of the direction which it indicates and the coordinate system under consideration.

14. Name 3 coordinate systems used in electromagnetic engineering?

1) Cartesian or rectangular coordinate system. 2) Cylindrical coordinate system. 3) Spherical coordinate system.

A point in rectangular coordinate system is located by three coordinates namely x, y and z coordinates. The point can be reached by moving from origin, the distance x in x direction then the distance y in y direction and finally z in z direction.

16. What is separation of vector?

The distance vector is also called as separation vector. Distance vector is nothing but the length of the vector.

17. State Distance formula?

Distance formula give the distance between the two points representing tips of the vector.

18. Show how a point p represented in a spherical coordinate system.

The point p can be defined as the intersection of three surfaces in spherical coordinate system. r - Constant which is a sphere with centre as origin θ - Constant which is a right circular cone with apex as origin and axis as z axis. Φ - Constant is a plane perpendicular to xy plane.

19. State the relationship between Cartesian and spherical system?

$x = r \sin \theta \cos \Phi$ $y = r \sin \theta \sin \Phi$ $z = r \cos \theta$ Now r can be expressed as $x^2 + y^2 + z^2 = r^2$
 $\sin^2 \theta [\cos^2 \Phi + \sin^2 \Phi] + r^2 \cos^2 \theta = r^2$
 $\sin^2 \theta [\cos^2 \Phi + \sin^2 \Phi] + r^2 \cos^2 \theta = r^2$
 $\sin^2 \theta + \cos^2 \theta = 1$

20. What are the types of integral related to electromagnetic theory?

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1. Line integral 2. Surface integral 3. Volume integral

21. Give the types of charge distribution.

1. Line charge 2. Point charge 3. Surface charge 4. Volume charge.

16 marks

1. (a) State and prove divergence theorem.

(b) What are the major sources of electromagnetic fields.

2. Check validity of the divergence theorem considering the field $D = 2xy \mathbf{a}_x + x^2 \mathbf{a}_y$ c/m² and the rectangular parallelepiped formed by the planes $x=0, x=1, y=0, y=2$ & $z=0, z=3$. 3. A vector field $D = [5r^2/4]\mathbf{I}_r$ is given in spherical co-ordinates. Evaluate both sides of divergence theorem for the volume enclosed between $r=1$ & $r=2$.

3. Explain three co-ordinate systems.

4. Obtain the expressions for differential area and volume element in cylindrical coordinate system. (13)

5. Analyze the geometrical position of the point in Cartesian coordinate system and obtain the algebraic equations. (13)

6. Express the space rate of change of a scalar in a given direction in terms of its gradient. (13)

7. Apply divergence theorem to find the divergence of the vector field in curvilinear coordinate system.

8. State and prove divergence theorem for a given differential volume element. (13)

9. Assess the position of the vector field in spherical coordinate system and derive the expressions for differential areas. (13)

10. Explain the difference between irrotational and solenoidal field using Helmholtz theorem. (13)

11. (i) Verify the null identities using general orthogonal curvilinear coordinates.

(ii) How do you transform the vectors between Cartesian and cylindrical systems? (6)

12. Elaborate the Stoke's theorem with their applications.

Problems

EC3452_ELECTROMAGNETIC FIELDS

<https://play.google.com/store/apps/details?id=info.therithal.brainkart.annauniversitynotes>

1. The three vertices of a triangle are located at $A(-1,2,5)$, $B(-4,-2,-3)$, and $C(1,3,-2)$.

(a) Find the length of the perimeter of the triangle.

(b) Find a unit vector that is directed from the midpoint of the side AB to the midpoint of the side BC.

(c) Show that this unit vector multiplied by a scalar is equal to the vector from A to C and that the unit vector is therefore parallel to AC. [12M]

2. The vector from the origin to point A is given as $(6,-2,-4)$, and the unit vector directed from the origin toward point B is $(2, -2,1)/3$. If points A and B are ten units apart, find the coordinates of point B. [12M] 3. A circle, centred at the origin with radius of 2 units, lies in the xy plane. Determine the unit vector in rectangular components that lies in the xy plane, is tangent to the circle at $(1, 0)$, and is in the general direction of increasing values of y. [12M]

4. A vector field is specified as $G = 24xy\mathbf{a}_x + 12(x^2 + 2y)\mathbf{a}_y + 18z^2\mathbf{a}_z$. Given two points $P(1,2,-1)$ and $Q(2,1,3)$, find: (a) G at P; (b) a unit vector in the direction of G at Q; (c) a unit vector directed from Q towards P; (d) the equation of surface on which $G = 60$. [12M]

5. A field is given as $G = [25/(x^2 + y^2)](x\mathbf{a}_x + y\mathbf{a}_y)$. Find: (a) a unit vector in the direction of G at $P(3,4,-2)$; (b) the angle between G and $\mathbf{a}_x$ at P; (c) the value of double integral on the plane $y=7$. [12M]

6. Three vectors extending from the origin are given as $\mathbf{r}_1 = (7,3,-2)$, $\mathbf{r}_2 = (-2,7,-3)$ and $\mathbf{r}_3 = (0,2,3)$. Find: (a) a unit vector perpendicular to both $\mathbf{r}_1$ and $\mathbf{r}_2$; (b) a unit vector perpendicular to the vectors $\mathbf{r}_1 - \mathbf{r}_2$ and $\mathbf{r}_2 - \mathbf{r}_3$; (c) The area of the triangle defined by $\mathbf{r}_1$ and $\mathbf{r}_2$; (d) The area of the triangle defined by the heads of $\mathbf{r}_1, \mathbf{r}_2$, and $\mathbf{r}_3$. [12M]

7. (a) Express the field $D = (x^2 + y^2)^{-1}(x\mathbf{a}_x + y\mathbf{a}_y)$ in cylindrical components and cylindrical variables; (b) Evaluate D at the point where $\rho=2$, $\Phi=0.2\pi$, and $z=5$, expressing the result in cylindrical and rectangular components. [12M]

8. Express in cylindrical components; (a) the vector from $C(3,2,-7)$ to $D(-1,-4,2)$; (b) a unit vector at D directed toward C; (c) a unit vector at D directed toward the origin. [12M]

9. The surfaces $\rho=3$, $\rho=5$, $\Phi=100^\circ$, $\Phi=130^\circ$, $z=3$, and $z=4.5$ define a closed surface. (a) Find enclosed volume; (b) Find the total area of enclosing surface; (c) Find the total length of the twelve edges of the surfaces; (d) Find the length of longest straight line that lies entirely within the volume. [12M]

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10. Given point $P(r=0.8, \theta=30^\circ, \Phi=45^\circ)$, and $E = \frac{1}{r^2} (\cos\Phi \mathbf{a}_r + \sin\Phi/\sin\theta \mathbf{a}_\Phi)$; (a) Find E at P ; (b) Find E at P ; (c) Find a unit vector in the direction of E at P . [12M]

Unit 2Part A - Two Marks

1. Define point charge.

A point charge means that electric charge which is separated on a surface or space whose geometrical dimensions are very small compared to other dimensions, in which the effect of electric field to be studied.

2. Define one coulomb.

One coulomb of charge is defined as the charge possessed by $(1/1.602 \times 10^{-19})$ i.e 6×10^{18} number of electrons.

3. What are the various types of charge distribution? Give an example for each.

1. Point charge - Ex. Positive charge
2. Line charge - Ex. A sharp beam in a cathode ray tube.
3. Surface charge - Ex. The plate of a charged parallel plate capacitor.
4. Volume charge - Ex. The charged cloud.

4. State the assumptions made while defining a Coulomb's law.

- 1) The two charges are stationary.
- 2) The two charges are point charge.

5. What is an equipotential surface?

An equipotential surface is an imaginary surface in an electric field of a given charge distribution, in which all points on the surface are at the same electric potential.

6. What is an electric flux?

The total number of lines of force in any particular electric field is called electric flux. It is represented by the symbol ψ . Similar to the charge, unit of electric flux is also Coulomb

7. Define electric flux density.

The net flux passing normal through the unit surface area is called electric flux density. It is denoted as D . It has a specified direction which is normal to the surface area under consideration hence it is a vector field.

8. State Gauss's Law.

The electric flux passing through any closed surface is equal to the total charge enclosed by that surface

9. State the application of Gauss's law.

- 1) The Gauss's law can be used to find E and D for symmetrical charge distributions.
- 2) It is used to find the charge enclosed or the flux passing through the closed surface.

10. What is an equipotential surface?

An equipotential surface is an imaginary surface in an electric field of a given charge distribution, in which all points on the surface are at the same electric potential.

11. Define the unit of Potential difference.

The unit of potential difference is Volt. One Volt potential difference is one Joule of work done in moving unit charge from one point to other in the field.

12. Define potential difference.

The work done per unit charge in moving unit charge from B to A in the field E is called potential difference between the points B to A.

13. Define relaxation time.

The relaxation time τ is defined as the time required by the charge density to decay to 36.8% of its initial value

14. What is Potential Gradient?

The rate of change of potential with respect to the distance is called potential gradient.

15. What is Gaussian surface? What are the conditions to be satisfied in special Gaussian surface?

The surface over which is the Gauss's law is applied is called Gaussian surface. Obviously such a surface is a closed surface and it has to satisfy the following conditions.

- 1) The surface may be irregular but should be sufficiently large so as to enclose the entire charge.
- 2) The surface must be closed.
- 3) At each point of the surface D is either normal or tangential to the surface.
- 4) The electric flux density D is constant over the surface at which D is normal.

16. What is Gradient of V?

The maximum value of rate of change of potential with distance dv/dL is called gradient of V

17. Define Absolute potential.

The work done in moving a unit charge from infinity (or from reference point at which potential is zero) to the point under the consideration against E is called absolute potential of that point.

18. What is Polarization?

The applied field E shifts the charges inside the dielectric to induce the electric dipoles. This process is called Polarization.

19. What is Polarization of Dielectrics?

Polarization of dielectric means, when an electron cloud has a centre separated from the nucleus. This forms an electric dipole. The dipole gets aligned with the applied field.

20. What is method of images?

The replacement of the actual problem with boundaries by an enlarged region or with image charges but no boundaries is called the method of images.

21. When is method of images used?

Method of images is used in solving problems of one or more point charges in the presence of boundary surfaces.

1. (a) State and explain Coulomb's law indicating clearly the units of quantities in the equation of force? 6M (b) State and prove Gauss's law and write limitations of Gauss's law? 6M

2. A charge Q_0 located at the origin in free space, produces a field for which $E_2 = 1 \text{ kV/m}$ at point P (-2, 1, -1). (a) Find Q_0 . Find E at M (1, 6, 5) in (b) Cartesian coordinates. (c) Cylindrical coordinates. 12M

3. Four positive point charges 10-12 coulomb each are situated in X-Y plane at points (0, 0), (0, 1) (1, 1) and (1, 0) m. Find the electric field and potential at (3/4, 3/4) and (1, 1)? 12M

4. (a) Derive the expression for the electric field intensity due to line charge? 6M (b) Four concentrated charges $Q_1 = 0.3 \mu\text{C}$, $Q_2 = 0.2 \mu\text{C}$, $Q_3 = -0.3 \mu\text{C}$, $Q_4 = 0.2 \mu\text{C}$ are located at the vertices of a plane rectangle. The length of rectangle is 5 cm and breadth of the rectangle is 2 cm. Find the magnitude and direction of resultant force on Q_1 ? 6M

5. (a) Derive Laplace and Poisson's equation? 6M (b) Find electric potential due to electric dipole? 6M

6. (a) Derive the expression for electric field intensity at a point due to electric dipole? 6M (b) Derive an expression for electric potential due to point charge? 6M

7. (a) Derive Maxwell first equation? 6M (b) Derive the expression for torque on electric dipole in the presence of uniform electric field? 6M

8. Four point charges each of $10 \mu\text{C}$ are placed in free space at the points (1, 0, 0), (-1, 0, 0), (0, 1, 0) and (0, -1, 0) m respectively. Determine the force on a point charge of $30 \mu\text{C}$ located at a point (0, 0, 1) m? 12M

9. Derive electrical field intensity due to charged circular ring? 12M

10. a) Write the condition for Laplace equation? 2M b) What is the relation between electric flux density and electric field intensity? 2M c) Define dipole moment? 2M d) Define an electric dipole? 2M e) State vector form of coulombs law?

11 (a) Derive the continuity equation. What is its physical significance? 6M (b) Derive the point form of ohms law? 6M 2. Explain the boundary conditions of two perfect dielectrics materials? 12M

12. Explain the boundary conditions between conductor and free space? 12M

13. Explain the phenomenon of polarization when a dielectric slab is subjected to an electric field? 6M
14. (a) Derive the expression for capacitance of the spherical condenser? 6M (b) Find the current in circular wire, if the current density is A/m^2 . The radius of the wire is 2mm?
15. (a) Derive the expression for capacitance of a co-axial cable? 6M (b) A parallel plate capacitor has a plate area of $1.5m^2$ and a plate separation of 5mm. There are two dielectrics in between the plates. The first dielectric has a thickness of 3mm with a relative permittivity of 6 and the second has a thickness of 2mm with a relative permittivity of 4. Find the capacitor? 6M
16. (a) Derive the expression for parallel plate capacitor? 6M (b) What is the energy stored in a capacitor made of two parallel metal plates each of 30 cm^2 area separated by 5mm in air. $\epsilon_0 = 8.854 \times 10^{-12}$. The capacitor is charged to potential difference of 500v? 6M
17. At the boundary between glass $\epsilon_r=4$ and air, the lines of electric field make an angle of 40° with normal to the boundary. If electric flux density in the air is $0.25\mu C/m^3$. Determine the orientation and magnitude of electric flux density in the glass? 12M
18. A parallel plate capacitor consists of two square metal plates with 500mm side and separated by 10mm. a slab of sulphur ($\epsilon_r= 4$) 6mm thick is placed on the lower plate and air gap of 4mm. find capacitance of capacitor? 12M
19. a) Define polarization in dielectric materials? 2M b) Write the relation between current I and current density J? 2M c) Write the equation for energy stored in capacitor? 2M d) Define Dielectric Strength? 2M e) A conductivity of a wire is 5000 mho/m and it is subjected to an electric field of 0.1 volts/m. Then what is the current density (J) in a wire?

Unit 3**2 marks****1. State Uniqueness Theorem.**

The Uniqueness theorem can be stated as, If the solutions of Laplace's equation satisfy the boundary condition then that solution is unique, by whatever method is obtained. The solution of Laplace's equation gives the field which is unique satisfying the same boundary conditions, in a given region.

2. State the applications of Poisson's equation and Laplace's equation. 1) To obtain potential distribution over the region. 2) To obtain E_r in the region. 3) To check whether given region is free of charge or not. 4) To obtain the charge induced on the surface of the region.

3. Define current density.

The current density is defined as the current passing through the unit surface area, when the surface is held normal to the direction of the current. The current density is measured in A/m^2 .

4. Define a current and its unit Ampere. The current is defined as the rate of flow of charge and is measured as Ampere's. A current of 1 Ampere is said to be flowing across the surface when the charge of 1 coulomb is passing across the surface in 1 second.

5. What is drift current and convection current? The current constituted due to the drifting of electrons in metallic conductor is called drift current. While in dielectrics, there can be flow of charges, under the influence of electric field intensity. Such a current is called convection current.

6. State the principle of conservation of charge. The principle of conservation of charge is, the charges can neither be created nor be destroyed.

7. What is drift velocity? Under the effect of applied electric field, the available free electrons start moving. The moving electrons strike the adjacent atoms and rebound in the random directions. This is called drifting of the electrons. After sometime, the electrons attain the constant average velocity called drift velocity.

8 Define the unit of Potential difference. The unit of potential difference is Volt. One Volt potential difference is one Joule of work done in moving unit charge from one point to other in the field. $E_r \text{ coulomb joule Volt } 1 \text{ V} = 1 \text{ J/C}$.

9. Define dielectric strength. The minimum value of the applied electric field at which the dielectric breaks down is called dielectric strength of dielectric.

10. What is Polarization? The applied field E_r shifts the charges inside the dielectric to induce the electric dipoles. This process is called Polarization.

11. Define potential difference.

The work done per unit charge in moving unit charge from B to A in the field E_r is called potential difference between the points B to A. $\int_B^A E_r \cdot dl$.

12. What is Polarization of Dielectrics?

Polarization of dielectric means, when an electron cloud has a centre separated from the nucleus. This forms an electric dipole. The dipole gets aligned with the applied field.

13. State the point form of Ohm's law

The relationship between J and E can also be expressed in terms of conductivity of the material. Thus for metallic conductor,

$$J = \sigma E$$

Where σ - conductivity of material. And the equation is called point form of Ohm's law.

14. What is Potential Gradient?

The rate of change of potential with respect to the distance is called potential gradient.

16 marks

1. Derive the expression for torque produced on a closed current carrying when placed in a magnetic field? 10M
2. Using Biot-savart's law. Find due conductor of finite length? 12M
3. Find the magnetic field intensity due to co-axial cable? 12M
4. (a) Write down Maxwell's third equation in point and integral form? 6M (b) Find magnetic field intensity due to solenoid carrying current I and having length $L=4m$? 6M
5. (a) State and explain Biot-savart's law? 6M (b) Explain Maxwell's second equation? 6M
6. (a) State and explain ampere's circuital law? 6M (b) Derive the expression for the force between two current carrying wires? 6M
7. (a) A circular loop is located on $X^2 + Y^2 = 9$ and $Z=0$ carries a direct current of 10A along direction. Determine at $(0, 0, 5) m$? 6M (b) Explain about Magnetic Dipole Moment? 6M
8. Using Biot-savart law or Amperes law find due conductor of infinite length? 12M

9. (a) Explain relationship between magnetic torque and moment? 6M (b) Derive an expression for the force between two current carrying wires? 6M
10. a). Define Magnetic dipole moment? 2M b). Write Lorentz force equation? 2M c) State point form of Amperes law? 2M d) State Ampere's circuital law? 2M e) State Biot-savats law? 2M
11. (a) What is vector magnetic potential? Derive vector poisson's equation? 6M (b) A toroid has air core and has a cross sectional area of 10mm^2 it has 1000 turns and its mean radius is 10mm. find its inductance? 6M
12. Derive the expression for self inductance of solenoid and toroid? 12M
13. (a) A coil of 500 turns is wound on a closed iron ring of mean radius 10cm and cross section of 3 cm^2 . Find the self inductance of the winding if the relative permeability of iron is 800? 6M (b) Explain the difference between self inductance and mutual inductance? 6M
14. (a) Derive the expression for inductance of a co-axial cable? 6M (b) Compute the energy density in free space on account of field having $= 1000\text{ A/m}$? 6M
15. (a) A coil of 1000 turns is wound on a Toroidal iron ring of mean radius 10cm and cross section of 3cm^2 . Find the self inductance of the winding if the relative permeability of iron is 800? 6M (b) Explain scalar magnetic potential and its limitations? 6M
16. Find the mutual inductance between a long, straight wire and square loop lying in same plane? 12M
17. Derive an expression for the force between two straight long and parallel conductors? 12M
18. Derive the expression for Neuman's formula? 12M
19. Explain the concept of energy stored in magnetic fields? 12M 10. a) Define self inductance? 2M b) Define mutual inductance? 2M c) Define inductance? 2M d) What is the inductance of Solenoid? 2M e) What is the inductance of Toroid? 2M
13. Derive an expression for motional and transformer induced emf? 12M
14. What is displacement current? Explain physical significance of displacement current?
15. Explain faradays law of electromagnetic induction and derive the expression for induced e.m.f?

Unit 4**2 marks**

1. Define Magnetic flux density. The total magnetic lines of force i.e. magnetic flux crossing a unit area in a plane at right angles to the direction of flux is called magnetic flux density. It is denoted as B_r . Unit Wb/m².

2. State Ampere's circuital law. The line integral of magnetic field intensity H_r around a closed path is exactly equal to the direct current enclosed by that path. The mathematical representation is

$$\oint H \, dL = I_r$$

3. Define Magnetic field Intensity. Magnetic Field intensity at any point in the magnetic field is defined as the force experienced by a unit north pole of one Weber strength, when placed at that point. Unit: N/Wb (or) AT /m. It is denoted as H_r .

4. What is rotational and irrotational vector field? If curl of a vector field exists then the field is called rotational. For irrotational vector field, the curl vanishes i.e. curl is zero.

5. State Stoke's Theorem. The line integral of a vector A_r around a closed path L is equal to the integral of curl of A_r over the open surface S enclosed by the closed path L .

6. Give the application of Stoke's theorem. The Stoke's theorem is applicable for the open surface enclosed by the given closed path. Any volume is a closed surface and hence application of Stoke's theorem to a closed surface which enclosed certain volume produces zero answer.

7. Define Inductance. In general, inductance is also referred as self inductance as the flux produced by the current flowing through the coil links with the coil itself.

8. What is fringing effect? If there is an air gap in between the path of the magnetic flux, it spreads and bulges out. This effect is called fringing effect.

9. What are boundary conditions? The conditions of the magnetic field existing at the magnetic field existing at the boundary of the two media when the magnetic field passes from one medium to other are called boundary conditions.

10. Define self inductance. Self inductance is defined as the rate of total magnetic flux linkage to the current through the coil.

11. State Biot Savart Law. The Biot Savart law states that, The magnetic field intensity dH_r produced at a point p due to a differential current element IdL is 1) Proportional to the product of the current I and differential length dL 2) The sine of the angle between the element and the line

joining point p to the element 3) And inversely proportional to the square of the distance R between point p and the element $2 \sin R \text{ IdL dH } \theta \propto r$.

12. What is Magnetostatics? The study of steady magnetic field, existing in a given space, produced due to the flow of direct current through a conductor is called Magnetostatics. 13. What is Magnetic Field? The region around a magnet within which influence of the magnet can be experienced is called Magnetic Field.

14. What are Magnetic Lines of Force? The existence of Magnetic Field can be experienced with the help of compass field. Such a field is represented by imaginary lines around the magnet which are called Magnetic Lines of Force.

15. Define Right hand Thumb Rule and where it is used? Right hand Thumb Rule states that, hold the current carrying conductor in the right hand such that the thumb pointing in the direction of current and parallel to the conductor, then curled fingers point in the direction of magnetic lines of flux around it. It is used to determine the direction of Magnetic field around a conductor carrying a direct current.

16. Define Right handed Screw Rule. It states that, imagine a right handed screw to be along the conductor carrying current with its axis parallel to the conductor and tip pointing in the direction of the current flow. Then the direction of Magnetic field is given by the direction in which screw must be turned so as to advance in the direction of current flow.

17. Give any four properties of Curl. 1. The Curl of a vector is a vector quantity. 2. $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$. 3. The curl of a scalar makes no sense. $\nabla \times \alpha = 0$ if α is scalar. 4. The Curl of gradient of a vector is Zero. $\nabla \times \nabla \phi = 0$

18. Give the relation between Magnetic flux and Flux density. The relation between Magnetic flux and flux density is obtained through the property of medium and permeability μ . This is given by, $\mathbf{B} = \mu \mathbf{H}$. 19. State Law of conservation of Magnetic Flux. It states that, the integral $\oint \mathbf{B} \cdot d\mathbf{l}$ over a closed surface is always zero. $\oint \mathbf{B} \cdot d\mathbf{l} = 0$. This is also called Gauss's law in integral form for magnetic fields.

20. Give Gauss's law in differential form for magnetic fields. The divergence of magnetic flux density is always zero. $\nabla \cdot \mathbf{B} = 0$.

21. Define scalar magnetic Potential. The scalar magnetic potential V_m can be defined for source free region where $\mathbf{J} = 0$ i.e. current density is zero.

22. Define Magneto static energy density. The magneto static energy density function is defined as $W_m = \frac{1}{2} \int \mathbf{H} \cdot d\mathbf{l} = \frac{1}{2} \int \mathbf{B} \cdot d\mathbf{l} = \frac{1}{2} \int \mu \mathbf{H}^2 dV$.

23. Define Mutual inductance. The mutual inductance between the two coils is defined as the ratio of flux linkage of one coil to the current in other coil. 24. State Kirchoff's Flux law. It states

that the total magnetic flux arriving at any junction in a magnetic circuit is equal to the magnetic flux leaving that junction. Using this law, parallel magnetic circuits can be easily analyzed. Mathematically, Kirchoff's flux law at a junction can be expressed as $\Sigma \phi = 0$.

16 marks

1. Illustrate the inconsistency of Ampere's circuital law for time varying fields, and what is the remedial solution proposed by Maxwell? Also give the differential form representation of all the four Maxwell's equations. [Understand] May 21
2. Derive wave equation from Maxwell's equation.
(6) [Understand] May 21
3. Write Faraday's law in differential and integral forms and explain Faraday's experiments. [Understand] May 21
4. Write Maxwell's equation in differential and integral forms. Examine them and give its physical interpretation. [Understand] 13 (May 19, 17R)
5. Derive wave equations for electric and magnetic fields. [Understand] 13 (May 19, 17R)
6. Derive the Maxwell's equation in differential and integral forms. [Understand] (13) Apr 18N, Nov 17N (16) (May 2016N)
7. With relevant practical applications explain in detail the particle applications of electromagnetic fields. [Understand] (15) (May 17N)
8. From the basic laws derive the time varying Maxwell's equation and explain the significance of each equation in detail. [Understand] (13) (May 17N, Nov 16N)
9. Explain the transformer emf using Faraday's law. [Understand] (5) (May 17N)
10. Starting from Maxwell's equation derive the equation for E field in the form of wave in free space. [Understand] (13) (May 17N)
11. Explain the condition and propagation of uniform plane wave in good conductor and derive the wave constants. [Understand] (13) (May 17N)
12. Derive the wave equations for electric and magnetic fields. [Understand] (16) (Nov 16N) (8) (Nov 15)
13. Starting from Maxwell's equation, derive homogeneous vector Helmholtz's equation in phasor form. [Understand] (16) (May 2016N)
14. Derive the Ampere's law. [Understand] (8) (May 2016)

15. With necessary explanation, derive the Maxwell's equation in differential and integral forms. [Understand] (16) (May 2016)

16. The conduction current flowing through a wire with conductivity $\sigma = 3 \times 10^7$ S/m and the relative permeability $\epsilon_r = 1$ is given by $I_c = 3 \sin \omega t$ (mA). $\omega = 10^8$ rad/sec, find the displacement current. [Apply] (8)(May 2016)

17. An electric field in a medium which is source free is given by $E = 1.5 \cos(10^8 t - \beta z) \hat{a}_x$ V/m. Find B, H and D. Assume $\epsilon_r = 1$, $\mu_r = 1$, $\sigma = 0$. [Apply] (8) (May 2016)

18. Derive Wave Equation from Maxwell's equations. [Understand] (8)(May 2016)

19. Derive the Maxwell's equation both in integral form and differential form Ampere's law and Gauss's law. [Understand] (16) (Nov 2015N)

20. State and explain the pointing theorem and derive the expression for pointing vector. [Understand] (16) (Nov 2015N)

21. Write the inconsistency of Ampere's law. Is it possible to construct generator of EMF which is constant and does not vary with time by using EM induction principle? Explain. [Understand] (16) (Nov 2015)

22. The electric field intensity of a linearly polarized uniform plane wave propagating in the +z direction in seawater is $E = 100 \cos(10^7 \pi t) \hat{i}$ V/m at $z=0$. The constitutive parameters of seawater are $\epsilon_r = 72$, $\mu_r = 1$, and conductivity $\sigma = 4$ S/m. Determine the attenuation constant, phase constant, intrinsic impedance, phase velocity, wavelength and skin depth. Also find the distance at which the amplitude of E is 1% of its value at $z=0$. [Apply] (8) (Nov 2015)

23. Is it possible to construct a generator of emf which is constant and does not vary with time by using the principle of EM inductor? Explain. [Analyse] (6) (May 15N)

24. In a parallel plate capacitor, a time-varying current $i(t) = I_m \cos \omega t$ flows through its leads. The plates have the surface area S and the distance between them is d. Show that the displacement current through the capacitor is exactly $I_m \cos \omega t$. Ignore the fringing effects. [Understand] (10) (May 15N)

25. If $D = 20x \hat{a}_x - 15y \hat{a}_y + kz \hat{a}_z$ $\mu\text{C}/\text{m}^2$, find the value of K to satisfy Maxwell's equations for region $\sigma=0$ and $\rho_v=0$. [Apply] (4) (May 15N)

26 If $H = (3x \cos(3+6y \sin \alpha)) \hat{a}_z$, find J if field are invariant with time. [Apply] (4)(May 15N)

27. Derive the expression for total power flow in a coaxial cable. [Understand] (8) (May 15N)

28. From basic principles, derive maxwells four equation in integral form and differential form. [Understand] (12) (May 15)

29. State the modified form of amperes circuital law. Why was it modified? Justify. [Understand] (4) (May 15)
30. Derive expression for Instantaneous, average and complex pointing vector.[Understand] (12) (May 15)
31. Interpret $E \times H$. [Understand] (4) (May 15)
32. Derive Maxwell's equations from basic principles[Understand]. (10) (Nov 14)
33. Derive the expression for power flow in a co-axial cable. [Understand] (6) (Nov 14)
34. Derive the expression for pointing vector. [Understand] (10) (Nov 14)
35. Why is Ampere's circuital law modified? How is it modified? Substantiate. [Understand] (6) (Nov 14)
36. Derive wave Equation from Maxwell's Equations. [Understand] (8) (Nov 14)
37. Explain ampere's circuit law. [Understand] (8)(May 2014)
38. State and prove pointing theorem. [Understand] (8)(May 14,Nov 2012), (10)(Nov 2011, May 2011,Nov 2009)
39. Derive Maxwell's four equations in integral differential form with their physical interpretation. [Understand] (8) (May2014) (16)(Nov 2009,May 2009,Nov 2008,May 2008,May 2007,Nov 2006)
40. State and prove pointing theorem. Write the expression for instantaneous, average, complex pointing vector. [Understand] (Nov 2013)

Unit 5**2 marks**

1. Define a wave. If a physical phenomenon that occurs at one place at a given time is reproduced at other places at later times, the time delay being proportional to the space separation from the first location then the group of phenomena constitutes a wave
2. Mention the properties of uniform plane wave. i) At every point in space, the electric field E and magnetic field H are perpendicular to each other. ii) The fields vary harmonically with time and at the same frequency everywhere in space.
3. Define intrinsic impedance or characteristic impedance. It is the ratio of electric field to magnetic field. or It is the ratio of square root of permeability to permittivity of medium.
4. Give the characteristic impedance of free space. 377ohms
5. Define propagation constant. Propagation constant is a complex number $\gamma = \alpha + j\beta$ where γ is propagation constant
6. Define skin depth It is defined as that depth in which the wave has been attenuated to $1/e$ or approximately 37% of its original value.
7. Define Poynting vector. The pointing vector is defined as rate of flow of energy of a wave as it propagates. $P = E \times H$
8. State Poyntings Theorem. The net power flowing out of a given volume is equal to the time rate of decrease of the the energy stored within the volume- conduction losses.
9. Give the difficulties in FDM. FDM is difficult to apply for problems involving irregular boundaries and non homogeneous material properties.
10. Explain the steps in finite element method. i) Discrimination of the solution region into elements. www.Vidarthiplus.in www.vidyarthiplus.in ii) Generation of equations for fields at each element iii) Assembly of all elements iv) Solution of the resulting system
11. State Maxwell's fourth equation. The net magnetic flux emerging through any closed surface is zero.
12. State Maxwell's Third equation The total electric displacement through the surface enclosing a volume is equal to the total charge within the volume.
13. State the principle of superposition of fields. The total electric field at a point is the algebraic sum of the individual electric field at that point.

14. Define pointing vector. The vector product of electric field intensity and magnetic field intensity at a point is a measure of the rate of energy flow per unit area at that point.

15. Give the formula to find potential at a point which is surrounded by four orthogonal points in FDM. $V_0 = \frac{1}{4}(V_1 + V_2 + V_3 + V_4)$

16. Give the formula to find potential at a point which is surrounded by six orthogonal points in FDM. $V_0 = \frac{1}{6}(V_1 + V_2 + V_3 + V_4 + V_5 + V_6)$

17. Define loss tangent. Loss tangent is the ratio of the magnitude of conduction current density to displacement current density of the medium.

18. Define reflection and transmission coefficients. Reflection coefficient is defined as the ratio of the magnitude of the reflected field to that of the incident field.

19. Define transmission coefficients. Transmission coefficient is defined as the ratio of the magnitude of the transmitted field to that of incident field.

20. What will happen when the wave is incident obliquely over dielectric – dielectric boundary? When a plane wave is incident obliquely on the surface of a perfect dielectric part of the energy is transmitted and part of it is reflected. But in this case the transmitted wave will be refracted, that is the direction of propagation is altered.

21. What is the fundamental difference between static electric and magnetic field lines? There is a fundamental difference between static electric and magnetic field lines. The tubes of electric flux originate and terminate on charges, whereas magnetic flux tubes are continuous.

22. What are uniform plane waves? Electromagnetic waves which consist of electric and magnetic fields that are perpendicular to each other and to the direction of propagation and are uniform in plane perpendicular to the direction of propagation are known as uniform plane waves.

23. What is the significant feature of wave propagation in an imperfect dielectric? The only significant feature of wave propagation in an imperfect dielectric compared to that in a perfect dielectric is the attenuation undergone by the wave.

24. What is the major drawback of finite difference method? The major drawback of finite difference method is its inability to handle curved boundaries accurately.

25. What is method of images? The replacement of the actual problem with boundaries by an enlarged region or with image charges but no boundaries is called the method of images.

26. When is method of images used? Method of images is used in solving problems of one or more point charges in the presence of boundary surfaces.

27. Define power density. The power density is defined as the ratio of power to unit area.

Power density=power/unit area.

28. What is called wave velocity? The velocity of propagation is called as wave velocity. It is denoted as v . $\mu \epsilon v = 1$. For free space it is denoted by c and its value is $3 \times 10^8 \text{ m/s}$.

29. What is called as intrinsic impedance? The ratio of amplitudes of E and H of the waves in either direction is called intrinsic impedance of the material in which wave is travelling. It is denoted by η .

30. Why dielectric medium is lossless dielectric. For perfect dielectric medium, both the fields E and H are in phase. Hence there is no attenuation. Hence there is no loss.

16 marks

1. What is uniform plane wave? Derive the relationship between E and H in a uniform plane wave. [Understand] (May 21)

2. Derive expressions for instantaneous, average and complex pointing vectors. [Understand] (May 21) 3. Analyze the wave behavior at boundaries under oblique incidence and derive the Brewster's angle. [Understand] (May 21)

4. A 1.8 KHz wave propagates in a medium characterized by $\mu_r = 1.6$, $\epsilon_r = 25$ and conductivity $\sigma = 2.5 \text{ S/m}$. The electric field intensity in the region is given by $E = 0.1 e^{-\alpha z} \cos(2\pi f t - \beta z) \hat{i} \text{ (V/m)}$. Determine the attenuation constant, propagation constant, intrinsic impedance, phase velocity, skin depth and wavelength of the wave. (Apply) 15 (May 19 17R)

5. Derive Poynting theorem. [Understand] 13 (May 19 17R)

6. Analyze the wave reflection and transmission at normal incidence at the boundary between two linear media. [Understand] 13 (May 19 17R)

7. Deduce the Poynting's theorem from Maxwell's equation and find the total time average power, crossing a given surface S . [Understand] (May 18N, Nov 2017N)

8. In a medium characterized by $\sigma = 0$, $\mu = \mu_0$, $\epsilon = 4\epsilon_0$, $E = 20 \sin(108 t - \beta z) \hat{a}_y \text{ V/m}$. Calculate B and H . (Nov 2017N)

9. State and prove pointing theorem. Write the expression for instantaneous, average and complex polynting vector. [Understand] (8) (May 17N, Nov 16N) (16) (Nov 2015)

10. What is a uniform plane wave? Derive the relation between E and H in a uniform plane wave. [Understand] (10) (May 15)

11. Describe the concept of Plane Wave propagation in good conductors. [Understand] (8) (Nov 14)

12. Explain with relevant expressions, the concept of reflection of plane waves by a perfect dielectric at both normal and oblique incidence. [Understand] (16) (Nov 14)
13. Analyze the wave behavior at boundaries under oblique incidence and derive the Brewster angle. [Analyse] (12) (May 2014)
14. Prove that a linearly polarized wave can be resolved into right hand circularly polarized wave and a left hand circularly polarized wave of equal amplitude. [Understand] (4) (May 2014)
15. Derive the wave equation for Electric and Magnetic fields. [Understand] (8) (May 2014)
16. Derive the wave equation from Maxwell's equations. Give the illustration for plane waves in Good conductors. [Understand] (Nov 2013)
17. Explain the parallel polarization with respect to oblique incidence. Discuss the types of polarization. [Understand] (Nov 2013)
18. Explain in detail on what happens when the wave is incident [Understand] (May 2013) i) Normally on perfect conductor (8) ii) Obliquely to the surface of perfect dielectrics. (8)
19. A plane sinusoidal electromagnetic wave traveling in space has $E_{\max} = 150 \mu \text{ V/m}$. i) Find the accompanying $H_{\max}$ ii) Propagation is in X direction and in H is oriented in Y direction. What is the direction of E? iii) Compute the average power transmitted. [Apply] (May 2013)
20. From Maxwell's equation, derive the electromagnetic wave equation in conducting medium for E and H fields. [Understand] (10) (Nov 2012, Nov 2011, May 2011, Nov 2010)
21. Calculate the attenuation constant and phase constant for a uniform plane with frequency 100 GHz in a conducting medium for which $\mu_r = 1$ and $\sigma = 58 \times 10^6 \text{ S/m}$. [Apply] (6) (Nov 2012, May 2011)
22. With reference electromagnetic waves, explain the following Linear, Circular & Elliptical Polarization [Understand] (3,3,2) (Nov 2012)
23. A plane wave is incident, normally on a perfect conductor. Derive the expression for standing wave. Find the location of nodes and antinodes in E & H fields. Sketch the standing wave pattern. [Understand] (8) (Nov 2012)
24. A uniform plane wave in a medium having $\sigma = 10^{-3} \text{ S/m}$, $\epsilon = 80\epsilon_0$ and $\mu = \mu_0$ is having a frequency of 10 KHz. (May 2012) [Apply] (i) Verify whether the medium is good conductor. (3) (ii) Calculate the following: (1) Attenuation constant (2) (2) Phase constant (2) (3) Propagation Constant (2) (4) Intrinsic impedance (2) (5) Wavelength (2) (6) Velocity of propagation (3)

25. A uniform plane wave in free space is normally incident on a dielectric having relative permittivity 4 and relative permeability 1. The electric field of incident wave is given by $E = E_0 e^{-jz} \hat{a}_x$ to z

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

EE3301 -ELECTROMAGNETIC FIELDS

CLASS/SEM: II EEE/III SEM

UNIT I

TWO MARKS

1. Define scalar field?

A field is a system in which a particular physical function has a value at each and every point in that region. The distribution of a scalar quantity with a defined position in a space is called scalar field.

Ex: Temperature of atmosphere.

2. Define Vector field?

If a quantity which is specified in a region to define a field is a vector then the corresponding field is called vector field.

3. Define scaling of a vector?

This is nothing but, multiplication of a scalar with a vector. Such a multiplication changes the magnitude of a vector but not the direction.

4. What are co-planar vector?

The vectors which lie in the same plane are called co-planar vectors.

5. What is an identical vector?

Two vectors are said to be identical if their difference is zero. Thus A and B are identical if $A - B = 0$, i.e., $A = B$. Such two vectors are also called as equal vectors.

6. Define base vectors?

The base vectors are the unit vectors which are strictly oriented along the directions of the coordinate axes of the given coordinate system.

7. What is a position vector?

Consider a point $p(x, y, z)$ in Cartesian coordinate system. Then the position vector of point p is represented by the distance of point p from the origin directed from origin to point. This is also called as radius vector.

8. Define scalar product of vectors?

The scalar of the two vectors A and B is denoted $A \cdot B$ and defined as the

product of the magnitude of A and magnitude of B and the cosine of angle between them.

$$A \cdot B = |A||B|\cos\theta_{AB}$$

9. Define Divergence.

Divergence is defined as the net outward flow of the flux per unit volume over a closed incremental surface.

10. State Divergence Theorem.

The integral of the normal component of any vector field over a closed surface is equal to the integral of the divergence of this vector field throughout the volume enclosed that closed surface.

11. Define curl of a vector.

The maximum circulation of F per unit area as area tends to zero whose direction is normal to the surface is called curl of F

$$\nabla \times F = \text{Curl of } F$$

12. State Stoke Theorem.

The line integral of F around a closed path L is equal to the integral of curl of F over the open surface S enclosed by the closed path L .

Mathematically it is expressed as

$$\oint_L F \cdot dL = \int_S (\nabla \times F) \cdot dS$$

Where dL -perimeter of total surface S .

13. What is physical significance of curl of a vector field?

Curl gives rate of rotation. $\text{Curl } F$ gives work done per unit area.

14. What is physical significance of divergence?

Divergence of current density gives net outflow of current per unit volume
Divergence of flux density gives net outflow per unit volume. In general, divergence of any field density gives net outflow of that field per unit volume.

15. State the conditions for a field to be a) solenoidal b) irrotational.

- a) Divergence of the field has to be zero.
- b) Curl of the field has to be zero.

16. Define scalar and vector quantity?

The scalar is a quantity whose value may be represented by a single real number which may be positive or negative.e.g, temperature, mass, volume, density

A quantity which has both a magnitude and a specified direction in space is called a vector.e.g.force, velocity, displacement,acceleration.

17. How to represent a vector.

A vector can be represented by a straight line with an arrow in a plane. The length of the segment is the magnitude of a vector while the arrow indicates the direction of a vector. $\vec{OA}$

18. What is a unit vector? What is its function while representing a vector?

A unit vector has a function to indicate the direction. Its magnitude is always

unity, irrespective of the direction which it indicates and the coordinate system under consideration.

19. Name 3 coordinate systems used in electromagnetic engineering?

- 1) Cartesian or rectangular coordinate system.
- 2) Cylindrical coordinate system.
- 3) Spherical coordinate system.

20. How to represent a point in a Cartesian system?

A point in rectangular coordinate system is located by three coordinates namely x, y and z coordinates. The point can be reached by moving from origin, the distance x in x direction then the distance y in y direction and finally z in z direction.

21. What is separation of vector?

The distance vector is also called as separation vector. Distance vector is nothing but the length of the vector.

22. State Distance formula?

Distance formula give the distance between the two points representing tips of the vector.

23. What are differential elements in Cartesian system?

$$d\vec{l} = dx\vec{a}_x + dy\vec{a}_y + dz\vec{a}_z$$

$$|d\vec{l}| = \sqrt{(dx)^2 + (dy)^2 + (dz)^2}$$

$$dv = dx dy dz$$

$$ds = ds a_n$$

24. What are the differential elements in cylindrical system?

dr-differential length in r direction

$r d\theta$ -differential length in θ direction

dz -differential length in z direction

$$d\vec{l} = dr\vec{a}_r + r d\theta\vec{a}_\theta + dz\vec{a}_z$$

$$|d\vec{l}| = \sqrt{(dr)^2 + (r d\theta)^2 + (dz)^2}$$

$$dv = r dr d\theta dz$$

25. What are the differential elements in spherical coordinate system?

dr-differential length in r direction

$r d\theta$ -differential length in θ direction

$r \sin\theta d\phi$ -differential length in ϕ direction

$$d\vec{l} = dr\vec{a}_r + r d\theta\vec{a}_\theta + r \sin\theta d\phi\vec{a}_\phi$$

$$|d\vec{l}| = \sqrt{(dr)^2 + (r d\theta)^2 + (r \sin\theta d\phi)^2}$$

$$dv = r \sin\theta dr d\theta d\phi$$

26. Which are the surfaces used to define the cylindrical coordinate system?

ds_r

= differential vector surface area normal to r direction

$$= r d\phi dz a_r$$

= differential vector surface area normal to ϕ direction

$$= dr dz a_\phi$$

ds_z = differential vector surface area normal to z direction

$$= r dr d\phi a_z$$

27. State the relation between Cartesian and cylindrical coordinate system?

$$x = r \cos \phi$$

$$y = r \sin \phi$$

$$z = z$$

28. Show how a point p represented in a spherical coordinate system.

The point p can be defined as the intersection of three surfaces in spherical coordinate system.

r - Constant which is a sphere with centre as origin

θ - Constant which is a right circular cone with apex as origin and axis as z axis.

Φ - Constant is a plane perpendicular to xy plane.

29. State the relationship between Cartesian and spherical system?

$$x = r \sin \theta \cos \Phi$$

$$y = r \sin \theta \sin \Phi$$

$$z = r \cos \theta$$

Now r can be expressed as

$$x^2 + y^2 + z^2 = r^2 \sin^2 \theta \cos^2 \Phi + r^2 \sin^2 \theta \sin^2 \Phi + r^2 \cos^2 \theta$$

$$= r^2 \sin^2 \theta [\sin^2 \Phi + \cos^2 \Phi] + r^2 \cos^2 \theta$$

$$= r^2 [\sin^2 \theta + \cos^2 \theta]$$

$$= r^2$$

30. What is dot product?

Dot product is also called as scalar product. It is defined as the product of the magnitude of A and magnitude of B and cosine of the smallest angle between them.

$$A \cdot B = |A| |B| \cos \theta_{AB}$$

31. State dot product properties.

1) It obeys commutative law. $A \cdot B = B \cdot A$

2) It obeys distributive law. $A \cdot (B + C) = A \cdot B + A \cdot C$

3) If the dot product with itself is performed the result is square of the magnitude

$$A \cdot A$$

4) Any unit vector dotted with itself is unity. $a_x.a_x = a_y.a_y = 1$

32. What is called as cross product?

Cross product is also called as vector product. It is defined as the product of the magnitude of A and magnitude of B and sine of the smallest angle between them.

$$A \times B = |A| |B| \sin \theta \vec{a}_n$$

33. State cross product properties.

1) Cross product is not cumulative

$$\text{i.e. } A \times B \neq B \times A$$

2) Reversing the order of vectors, reverse its direction.

$$A \times B = -|B| |A|$$

34. Give the application of dot products.

1) To determine the angle between the two vectors,

$$\theta = \cos^{-1} \frac{A \cdot B}{|A| |B|}$$

2) To find the component of a vector in a given direction.

35. Give the application of cross product.

1) The cross product is used to determine the direction of force.

$$F = IL \times B$$

2) Another physical quantity which can be represented by cross product is moment of force.

$$M = \vec{r} \times F = |r| |F| \sin \theta \vec{a}_n$$

36. Define scalar triple product.

The scalar triple product is

$$A \cdot (B \times C) = B \cdot (C \times A) = C \cdot (A \times B)$$

37. State scalar triple product properties.

1) The scalar triple product is distributive.

2) If two of the three vectors are equal then the result of the scalar triple product is zero.

$$A \cdot (A \times C) = 0$$

38. Define vector triple product.

The vector triple product of the three vectors A, B, C are mathematically defined

as,

$$A \times (B \times C) = B(A.C) - C(A.B)$$

39. State vector triple product properties.

The vector triple product properties are

1. $B \times (C \times A) = C(B.A) - A(B.C)$

$$C \times (A \times B) = A(C.B) - B(C.A)$$

This is because dot product is commutative.

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2. $(A.B)C \neq A(B.C)$ And

$(A.B)C = C(A.B)$

This is because $A.B$ is a scalar and multiplication by scalar to a vector is cumulative.

40. Convert Cartesian to cylindrical system.

$$\begin{bmatrix} A_r \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

41. Transform the Cartesian system into spherical system.

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin\theta \cos\phi & \sin\theta \sin\phi & \cos\theta \\ \cos\theta \cos\phi & \cos\theta \sin\phi & -\sin\theta \\ -\sin\phi & \cos\phi & 0 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

42. Transform the cylindrical system into Cartesian system.

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_r \\ A_\phi \\ A_z \end{bmatrix}$$

43. Give the distance formula in coordinate system.

$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$ -Cartesian

$d = \sqrt{r_2^2 + r_1^2 - 2r_1r_2 \cos(\phi_2 - \phi_1) + (z_2 - z_1)^2}$ -Cylindrical

$d = \sqrt{r_2^2 + r_1^2 - 2r_1r_2 \cos\theta_2 \cos\theta_1 - 2r_1r_2 \sin\theta_1 \sin\theta_2 \cos(\phi_2 - \phi_1)}$ -Spherical

44. What are the types of integral related to electromagnetic theory?

1. Line integral
2. Surface integral
3. Volume integral

45. Give the curl vector of the Cartesian system.

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

46. Give the curl vector of cylindrical coordinate system.

$$\nabla \times \vec{F} = \frac{1}{r} \begin{vmatrix} \vec{a}_r & r\vec{a}_\phi & \vec{a}_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ F_r & F_\phi & F_z \end{vmatrix}$$

47. Give the curl vector if spherical coordinate system.

$$\nabla \times \vec{F} = \frac{1}{r \sin \theta} \begin{vmatrix} \vec{a}_r & r\vec{a}_\theta & r \sin \theta \vec{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ F_r & rF_\theta & r \sin \theta F_\phi \end{vmatrix}$$

48. Given two points A (5, 4, 3) and (2, 3, 4). Find mid point of AB.

$$\begin{aligned} \text{Mid point of AB} &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right) \\ &= \left(\frac{5 + 2}{2}, \frac{4 + 3}{2}, \frac{3 + 4}{2} \right) \\ &= (3.5, 3.5, 3.5) \end{aligned}$$

49. Given two points in Cartesian coordinate system as A (3, -2, 1), B (-3, -3, 5). Find distance from B to A.

$$\vec{BA} = \vec{A} - \vec{B} = [3 - (-3)]\vec{a}_x + [(-2) - (-3)]\vec{a}_y + [1 - 5]\vec{a}_z$$

$$= 6\vec{a}_x + \vec{a}_y - 4\vec{a}_z$$

$$|\vec{BA}| = \sqrt{(6)^2 + (1)^2 + (-4)^2} = 7.2801$$

50. Give the types of charge distribution.

1. Line charge
2. Point charge
3. Surface charge
4. Volume charge.

51. Define point charge.

A point charge means that electric charge which is separated on a surface or space whose geometrical dimensions are very very small compared to other dimensions, in which the effect of electric field to be studied.

52. Define one coulomb.

One coulomb of charge is defined as the charge possessed by $(1/1.602 \times 10^{-19})$ i.e 6×10^{18} number of electrons.

53. State Coulomb's law.

The coulomb's law states that force between the two point charges Q1 and Q2

- i) Acts along the line joining the two point charges
- ii) is directly proportional to the product of the charges
- iii) is inversely proportional to the square of the distance between them.

$$F \propto \frac{Q_1 Q_2}{R^2}$$

54. Define constant of proportionality (K).

It is defined as $k = \frac{1}{4\pi \epsilon}$ where ϵ is the permittivity of medium in which charges are

located, where $\epsilon = \epsilon_0 \epsilon_r$

55. What are the various types of charge distribution? Give an example for each.

- 1. Point charge-Ex. Positive charge
- 2. Line charge -Ex. A sharp beam in a cathode ray tube.
- 3. Surface charge-Ex. The plate of a charged parallel plate capacitor.
- 4. Volume charge-Ex. The charged cloud.

56. State the assumptions made while defining a Coulomb's law.

- 1) The two charges are stationary.
- 2) The two charges are point charge.

57. Define surface and volume charge density.

$$\sigma = \frac{\text{total charge in coulomb}}{\text{total area in sq. meters}} \text{ C/m}^2$$

$$\rho_v = \frac{\text{Total charge in coulomb}}{\text{Total volume in cubic meters}} \text{ C/m}^3$$

58. Define electric field intensity at a point.

The electric field intensity is defined as force exerted per unit charge.

$$E = \frac{Q_1}{4\pi \epsilon_0 R_1^2} \vec{a}_{1p}$$

Where p-position of any other charge around Q1

UNIT II

TWO MARKS

1. Define point charge.

A point charge means that electric charge which is separated on a surface or space whose geometrical dimensions are very very small compared to other dimensions, in which the effect of electric field to be studied.

2. Define one coulomb.

One coulomb of charge is defined as the charge possessed by $(1/1.602 \times 10^{-9})$ i.e 6×10^{18} number of electrons.

3. State Coulomb's law.

The coulomb's law states that force between the two point charges Q_1 and Q_2

- i) Acts along the line joining the two point charges
- ii) is directly proportional to the product of the charges
- iii) is inversely proportional to the square of the distance between them.

$$F \propto \frac{Q_1 Q_2}{R^2}$$

4. Define constant of proportionality (K).

It is defined as $k = \frac{1}{4\pi \epsilon}$ where ϵ is the permittivity of medium in which charges are

located. where $\epsilon = \epsilon_0 \epsilon_r$

5. Define electric field intensity at a point.

The electric field intensity is defined as force exerted per unit charge.

$$E = \frac{Q_1}{4\pi \epsilon_0 R_1^2} \vec{a}_{1p}$$

Where p-position of any other charge around Q_1

6. What are the various types of charge distribution? Give an example for each.

- 1. Point charge-Ex. Positive charge
- 2. Line charge -Ex. A sharp beam in a cathode ray tube.
- 3. Surface charge-Ex. The plate of a charged parallel plate capacitor.
- 4. Volume charge-Ex. The charged cloud.

7. State the assumptions made while defining a Coulomb's law.

- 1) The two charges are stationary.
- 2) The two charges are point charge.

8. Define surface charge density.

$$\rho_s = \frac{\text{total charge in coulomb}}{\text{total area in sq. meters}} \text{ C/m}^2$$

9. State the expression for electric field intensity due to infinite line charge.

$$E = \frac{\rho_L}{2\pi \epsilon_0 r} \vec{a}_r \text{ V/m}$$

Where r - perpendicular distance of point p from the line charge
 $\vec{a}_r$ - unit vector in the direction of the perpendicular distance of point p from the line charge.

10. State the expression for electric field intensity due to charged circular ring.

$$E = \frac{\rho_L r z}{2 \epsilon_0 (r^2 + z^2)^{3/2}} \vec{a}_z$$

Where z - perpendicular distance of point p from the ring along the axis of the ring.

11. Define conservative field.

The $\oint$ sign indicates integral over a closed path. Such a field having property of $\oint E \cdot dL = 0$ associated with it is called conservative field or lamellar field.

12. What is an equipotential surface?

An equipotential surface is an imaginary surface in an electric field of a given charge distribution, in which all points on the surface are at the same electric potential.

13. What is an electric flux?

The total number of lines of force in any particular electric field is called electric flux. It is represented by the symbol Ψ . Similar to the charge, unit of electric flux is also Coulomb.

14. Define electric flux density.

The net flux passing normal through the unit surface area is called electric flux density. It is denoted as D . It has a specified direction which is normal to the surface area under consideration hence it is a vector field.

15. State the vector form of electric flux density.

The flux density $\vec{D}$ at the point p can be represented in the vector form as

$$\vec{D} = \frac{d\Psi}{ds} \vec{a}_n \text{ C/m}^2.$$

Where d - total flux lines crossing normal through the differential surface ds .

$\bar{a}_n$ -unit vector in the direction normal to the differential surface area.

16. Derive the relationship between D and E .

$$\begin{aligned} E &= \frac{Q}{4\pi \epsilon_0 r^2} \bar{a}_r \\ \rightarrow \frac{Q}{4\pi r^2} \bar{a}_r \\ \bar{D} &= \epsilon_0 \bar{E} \\ \bar{D} &= \epsilon_0 \bar{E} \text{ for free space} \\ \bar{D} &= \epsilon \bar{E} \text{ in general} \end{aligned}$$

17. State Gauss's Law.

The electric flux passing through any closed surface is equal to the total charge enclosed by that surface.

18. State the mathematical representation of Gauss's law.

The common form used to represent Gauss's law mathematically is

$$\psi = Q = \oint_S \vec{D} \cdot d\vec{s} = \oint_V \rho_v \cdot dv.$$

19. State the application of Gauss's law.

1) The Gauss's law can be used to find E and D for symmetrical charge distributions.

2) It is used to find the charge enclosed or the flux passing through the closed surface.

20. What is Divergence? Explain its physical meaning.

The mathematical operation on A is called a divergence. It is denoted as divergence A . Hence mathematically divergence is given by

$$\text{div} \vec{A} = \lim_{\Delta V \rightarrow 0} \frac{\int_{\Delta V} \vec{A} \cdot d\vec{s}}{\Delta V} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}.$$

The physical meaning is, Let A be the flux density vector then, the divergence of the vector flux density A is the outflow of flux from a small closed surface per unit volume as the volume shrinking to zero.

21. State the Divergence theorem.

The integral of the normal component of any vector field over a closed surface is equal to the integral of the divergence of this vector field throughout the volume enclosed by that closed surface.

22. What is an equipotential surface?

An equipotential surface is an imaginary surface in an electric field of a given charge distribution, in which all points on the surface are at the same electric potential.

23. What is the formula for coaxial cable using gauss law?

$$\vec{E} = \frac{\rho L}{2\pi \epsilon_0 r} \vec{a}_r V / m$$

24. What is the formula for spherical shell of charge using gauss law?

For $r < a$, $\vec{E} = 0$

For $r = a$, $\vec{E} = Q / (4\pi\epsilon a^2) \vec{a}_r$

For $r > a$, $\vec{E} = Q / (4\pi\epsilon r^2) \vec{a}_r$

25. What is the formula for uniformly charged sphere using gauss law?

$r > a$, $\vec{D} = a^3 R / 3r^2 \vec{a}_r$

$r = a$, $\vec{D} = \rho_v a / 3 \vec{a}_r$

$r < a$, $\vec{D} = r / 3 \rho_v \vec{a}_r$

26. Define surface and volume charge density.

$$\rho_v = \frac{\text{Total charge in coulomb}}{\text{Total volume in cubic meters}} \text{ C/m}^3$$

27. Define the unit of Potential difference.

The unit of potential difference is Volt. One Volt potential difference is one Joule of work done in moving unit charge from one point to other in the field E .

$$1 \text{ Volt} = \frac{1 \text{ joule}}{1 \text{ coulomb}}$$

28. Define potential difference.

The work done per unit charge in moving unit charge from B to A in the field E is called potential difference between the points B to A.

$$V = - \int_B^A \vec{E} \cdot d\vec{l}$$

29. Define relaxation time.

The relaxation time is defined as the time required by the charge density to decay to 36.8% of its initial value.

$$\tau = \text{Relaxation Time} = \frac{\epsilon}{\sigma} \text{ sec.}$$

30. What is Potential Gradient?

The rate of change of potential with respect to the distance is called potential gradient.

$$\frac{dv}{dL} = \lim_{\Delta v \rightarrow 0} \frac{\Delta v}{\Delta L} = \text{Potential gradient.}$$

31. What is Gaussian surface? What are the conditions to be satisfied in special Gaussian surface?

The surface over which Gauss's law is applied is called Gaussian surface.

Obviously such a surface is a closed surface and it has to satisfy the following conditions.

- 1) The surface may be irregular but should be sufficiently large so as to enclose the entire charge.
- 2) The surface must be closed.
- 3) At each point of the surface $\vec{D}$ is either normal or tangential to the surface.
- 4) The electric flux density D is constant over the surface at which $\vec{D}$ is normal.

32. What is Gradient of V?

The maximum value of rate of change of potential with distance dv/dL is called gradient of V.

33. Define Absolute potential.

The work done in moving a unit charge from infinity (or from reference point at which potential is zero) to the point under the consideration against E is called absolute potential of that point.

$$V = \frac{Q}{4\pi \epsilon_0 r} \text{ volt}$$

34. What is the relationship between V and E?

$$E = -\text{grad}V$$

35. Define line charge density.

$$\rho_L = \frac{\text{total charge in coulomb}}{\text{total length}} \text{ C/m}$$

36. Write the formula for electric field intensity due to infinite line charge using Gauss law.

$$\vec{E} = \frac{\rho_L}{2\pi \epsilon_0 r} \vec{a}_r \text{ V/m}$$

The value of electric flux density is zero

38. What is dot product?

Dot product is also called as scalar product. It is defined as the product of the magnitude of A and magnitude of B and cosine of the smallest angle between them.

$$A \cdot B = |A| |B| \cos \theta_{AB}$$

39. What is physical significance of divergence?

Divergence of current density gives net outflow of current per unit volume. Divergence of flux density gives net outflow per unit volume. In general, divergence of any field density gives net outflow of that field per unit volume.

40. Write the formula for electric field intensity due to infinite sheet of charge using gauss law.

$$\vec{E} = (\rho_s / 2\epsilon_0) \vec{a}_z \text{ v/m}$$

UNIT III
TWO MARKS

1. State Poisson's equation.

The Poisson's equation is stated as,

$$\Delta^2 v = -\frac{\rho_v}{\epsilon}$$

2. State Laplace's equation in Cartesian system.

$$\Delta^2 v = \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0.$$

This is Laplace equation in Cartesian form.

3. State Laplace equation in cylindrical form.

$$\Delta^2 v = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 v}{\partial \phi^2} + \frac{\partial^2 v}{\partial z^2} = 0.$$

This is Laplace equation in cylindrical form.

4. State Laplace equation in Spherical form.

$$\Delta^2 v = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial v}{\partial \theta} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial^2 v}{\partial \phi^2} = 0.$$

This is Laplace equation in spherical system.

5. State Uniqueness Theorem.

The Uniqueness theorem can be stated as,

If the solutions of Laplace's equation satisfy the boundary condition then that solution is unique, by whatever method is obtained.

The solution of Laplace's equation gives the field which is unique satisfying the same boundary conditions, in a given region.

6. State the applications of Poisson's equation and Laplace's equation.

- 1) To obtain potential distribution over the region.
- 2) To obtain E in the region.
- 3) To check whether given region is free of charge or not.
- 4) To obtain the charge induced on the surface of the region.

7. Define current density.

The current density is defined as the current passing through the unit surface area, when the surface is held normal to the direction of the current. The current density is measured in A/m^2 .

8. Define a current and its unit Ampere.

The current is defined as the rate of flow of charge and is measured as Ampere's.

A current of 1 Ampere is said to be flowing across the surface when the charge of 1 coulomb is passing across the surface in 1 second.

9. What is drift current and convection current?

The current constituted due to the drifting of electrons in metallic conductor is called drift current.

While in dielectrics, there can be flow of charges, under the influence of electric field intensity. Such a current is called convection current.

10. State the principle of conservation of charge.

The principle of conservation of charge is, the charges can neither be created nor be destroyed.

11. What is drift velocity?

Under the effect of applied electric field, the available free electrons start moving. The moving electrons strike the adjacent atoms and rebound in the random directions. This is called drifting of the electrons. After sometime, the electrons attain the constant average velocity called drift velocity.

12. Define the unit of Potential difference.

The unit of potential difference is Volt. One Volt potential difference is one Joule of work done in moving unit charge from one point to other in the field E .

$$1\text{ Volt} = \frac{1\text{ joule}}{1\text{ coulomb}}.$$

13. Define dielectric strength.

The minimum value of the applied electric field at which the dielectric breaks down is called dielectric strength of dielectric.

14. What is Polarization?

The applied field E shifts the charges inside the dielectric to induce the electric dipoles. This process is called Polarization.

15. Define potential difference.

The work done per unit charge in moving unit charge from B to A in the field E is called potential difference between the points B to A.

$$V = -\int_B^A \vec{E} \cdot d\vec{l}.$$

16. State Law of refraction.

The law of refraction is stated as , $\frac{\tan\theta_1}{\tan\theta_2} = \frac{\epsilon_1}{\epsilon_2} = \frac{\epsilon_{r1}}{\epsilon_{r2}}$,

Where the angles θ_1 and θ_2 are dependent on permittivity of two media and not on $\vec{D}$ and $\vec{E}$.

17. Define relaxation time.

The relaxation time is defined as the time required by the charge density to decay to 36.8% of its initial value.

$$\tau = \text{Relaxation Time} = \frac{\epsilon}{\sigma} \text{ sec.}$$

18. What is Polarization of Dielectrics?

Polarization of dielectric means, when an electron cloud has a centre separated from the nucleus. This forms an electric dipole. The dipole gets aligned with the applied field.

19. State the point form of Ohm's law.

The relationship between $\vec{J}$ and $\vec{E}$ can also be expressed in terms of conductivity of the material. Thus for metallic conductor,

$$\vec{J} = \sigma \vec{E}$$

Where σ - conductivity of material. And the equation is called point form of Ohm's law.

20. What is Potential Gradient?

The rate of change of potential with respect to the distance is called potential gradient.

$$\frac{dv}{dL} = \lim_{\Delta v \rightarrow 0} \frac{\Delta v}{\Delta L} = \text{Potential gradient.}$$

21. What is Boundary conditions means?

The conditions existing at the boundary of the two media when field passes from one medium to other are called boundary conditions.

22. What is Gaussian surface? What are the conditions to be satisfied in special Gaussian surface?

The surface over which is the Gauss's law is applied is called Gaussian surface. Obviously such a surface is a closed surface and it has to satisfy the following conditions.

- 5) The surface may be irregular but should be sufficiently large so as to enclose the entire charge.
- 6) The surface must be closed.
- 7) At each point of the surface $\vec{D}$ is either normal or tangential to the surface.
- 8) The electric flux density D is constant over the surface at which $\vec{D}$ is normal.

23. What is Gradient of V?

The maximum value of rate of change of potential with distance dv/dL is called gradient of V.

24. How is electric energy stored in a capacitor?

In a capacitor, the work done in charging a capacitor is stored in the form of electric energy.

25. What are dielectrics?

Dielectrics are materials that may not conduct electricity through it but on applying electric field induced charges are produced on its faces. The valence electron in atoms of a dielectric are tightly bound to their nucleus.

26. What is a capacitor?

A capacitor is an electrical device composed of two conductors which are separated through a dielectric medium and which can store equal and opposite charges independent of whether other conductors in the system are charged or not.

27. What are the factors does the capacitance depends on?

1. The permittivity of the dielectric used.
2. The area of cross section of the plates
3. The distance of separation of the plates

28. Write the expression for parallel plate capacitor?

$$C = \epsilon_0 \epsilon_r A / d \text{ F}$$

29. Write the expression for capacitance of a coaxial cable?

$$C = (2\pi\epsilon L) / \ln (b/a) \text{ F}$$

30. Write the expression for capacitance of single isolated sphere?

$$C = 4\pi\epsilon a \text{ F}$$

31. What is meant by displacement current?

Displacement current is nothing but the current flows through the capacitor

32. What is the energy stored in a capacitor?

$$W = \frac{1}{2} cv^2 \text{ J}$$

33. Write the expression for spherical capacitance?

$$C = (4\pi\epsilon)/(1/a - 1/b) \text{ F}$$

34. Write the expression for isolated spherical conductor coated with dielectric?

$$C = 4\pi/(1/\epsilon_1(1/a - 1/r_1) + 1/\epsilon_0 r_1) \text{ F}$$

35. Write the expression for dielectric boundary normal to plates?

$$C = \epsilon_1 A_1/d + \epsilon_2 A_2/d \text{ F}$$

36. Write the expression for dielectric boundary parallel to plates?

$$C = A/(d_1/\epsilon_1 + d_2/\epsilon_2 + \dots) \text{ F}$$

37. What is meant by multiple dielectric capacitors?

The multiple dielectric capacitor is one in which the space between the plates is filled with more than one dielectrics

38. What are the two situations of the boundary conditions based on nature of the media?

1. Boundary between conductor and free space.
2. Boundary between two dielectrics with different properties.

39. What meaning would you give to the capacitance of a single conductor?

Single conductors also possess capacitance. It is a capacitor whose one plate is at Infinity.

40. Define dielectric strength of a dielectric?

The minimum value of the applied electric field at which the dielectric breaks down is called dielectric strength of that dielectric

UNIT-IV
TWO MARKS

1. Define Magnetic flux density.

The total magnetic lines of force i.e. magnetic flux crossing a unit area in a plane at right angles to the direction of flux is called magnetic flux density. It is denoted as B . Unit Wb/m^2 .

2. State Ampere's circuital law.

The line integral of magnetic field intensity H around a closed path is exactly equal to the direct current enclosed by that path.

The mathematical representation is $\oint H \cdot dL = I$.

3. Define Magnetic field Intensity.

Magnetic Field intensity at any point in the magnetic field is defined as the force experienced by a unit north pole of one Weber strength, when placed at that point. Unit: N/Wb (or) AT /m . It is denoted as H .

4. What is rotational and irrotational vector field?

If curl of a vector field exists then the field is called rotational. For irrotational vector field, the curl vanishes i.e. curl is zero.

5. State Stoke's Theorem.

The line integral of a vector A around a closed path L is equal to the integral of curl of A over the open surface S enclosed by the closed path L .

6. Give the application of Stoke's theorem.

The Stoke's theorem is applicable for the open surface enclosed by the given closed path. Any volume is a closed surface and hence application of Stoke's theorem to a closed surface which enclosed certain volume produces zero answer.

7. Define Inductance.

In general, inductance is also referred as self inductance as the flux produced by the current flowing through the coil links with the coil itself.

8. What is fringing effect?

If there is an air gap in between the path of the magnetic flux, it spreads and bulges out. This effect is called fringing effect.

9. What are boundary conditions?

The conditions of the magnetic field existing at the magnetic field existing at the boundary of the

two media when the magnetic field passes from one medium to other are called boundary conditions.

10. Define self inductance.

Self inductance is defined as the rate of total magnetic flux linkage to the current through the coil.

11. State Biot Savart Law.

The Biot Savart law states that,

The magnetic field intensity dH produced at a point p due to a differential current element IdL is

- 1) Proportional to the product of the current I and differential length dL
- 2) The sine of the angle between the element and the line joining point p to the element
- 3) And inversely proportional to the square of the distance R between point p and the element

$$dH \propto \frac{IdL \sin \theta}{R^2}$$

12. What is Magnetostatics?

The study of steady magnetic field, existing in a given space, produced due to the flow of direct current through a conductor is called Magnetostatics.

13. What is Magnetic Field?

The region around a magnet within which influence of the magnet can be experienced is called Magnetic Field.

14. What are Magnetic Lines of Force?

The existence of Magnetic Field can be experienced with the help of compass field. Such a field is represented by imaginary lines around the magnet which are called Magnetic Lines of Force.

15. Define Right hand Thumb Rule and where it is used?

Right hand Thumb Rule states that, hold the current carrying conductor in the right hand such that the thumb pointing in the direction of current and parallel to the conductor, then curled fingers point in the direction of magnetic lines of flux around it. It is used to determine the direction of Magnetic field around a conductor carrying a direct current.

16. Define Right handed Screw Rule.

It states that, imagine a right handed screw to be along the conductor carrying current with its axis parallel to the conductor and tip pointing in the direction of the current flow. Then the direction of Magnetic field is given by the direction in which screw must be turned so as to advance in the direction of current flow.

17. Give any four properties of Curl.

1. The Curl of a vector is a vector quantity.
2. $\nabla \times (A + B) = \nabla \times A + \nabla \times B$.
3. The curl of a scalar makes no sense.
i.e. $\nabla \times \alpha = \text{No Sense}$ if α is scalar.
4. The Curl of gradient of a vector is Zero.
 $\nabla \times \nabla V = 0$

18. Give the relation between Magnetic flux and Flux density.

The relation between Magnetic flux and flux density is obtained through the property of medium and permeability μ . This is given by,

$$B = \mu H.$$

19. State Law of conservation of Magnetic Flux.

It states that, the integral $\oint B \cdot d\vec{s}$ over a closed surface is always zero.

$$\oint B \cdot d\vec{s} = 0.$$

This is also called Gauss's law in integral form for magnetic fields.

20. Give Gauss's law in differential form for magnetic fields.

The divergence of magnetic flux density is always zero.

$$\nabla \cdot B = 0.$$

21. Define scalar magnetic Potential.

The scalar magnetic potential V_m can be defined for source free region where J i.e. current density is zero.

22. Define Magneto static energy density.

The magneto static energy density function is defined as

$$W = \lim_{\Delta v \rightarrow 0} \frac{\Delta w_m}{\Delta v} = \frac{1}{2} \mu H^2.$$

23. Define Mutual inductance.

The mutual inductance between the two coils is defined as the ratio of flux linkage of one coil to the current in other coil. Thus the mutual inductance between circuit 1 and circuit 2 is given by

$$\mu_{12} = \frac{N_2 \Phi_{12}}{I_1} H.$$

24. State Kirchoff's Flux law.

It states that the total magnetic flux arriving at any junction in a magnetic circuit is equal to the

magnetic flux leaving that junction. Using this law, parallel magnetic circuits can be easily analyzed. Mathematically, Kirchoff's flux law at a junction can be expressed as

$$\Sigma \phi = 0.$$

25. State Kirchoff's MMF law.

Kirchoff's MMF law states that the resultant mmf around a closed magnetic circuit is equal to the algebraic sum of products of flux and reluctance of each part of the closed circuit. For closed magnetic circuit,

$$\Sigma MMF = \Sigma \phi R.$$

26. What is Magnetization?

The field produced due to the movement of bound charges is called Magnetization represented by M .

27. Define Reluctance.

Reluctance R is defined as the ratio of the magneto motive force to the total flux.

$$R = \frac{em}{\phi} \text{ And it is measured as Ampere-turn/Weber.}$$

28. What is Lorentz force equation?

Lorentz force equation relates mechanical force to the electrical force. It is given as the total force on a moving charge in the presence of both electric and magnetic fields.

$$F = F_e + F_m N.$$

29. Define Moment of force.

The Moment of a force or torque about a specified point is defined as the vector product of the moment arm R and the force F . It is measured in Nm.

$$T = R \times F Nm.$$

30. Define Magnetic dipole moment.

The Magnetic dipole moment of a current loop is defined as the product of current through the loop and the area of the loop, directed normal to the current loop.

31. Give any two dissimilarities between electric and magnetic circuits.

1) In electric circuit the current actually flows i.e. there is a movement of electrons whereas in magnetic circuit, due to m.m.f, flux gets established and doesn't flow in the sense in which current flows.

2) The electric lines of flux are not closed. They start from positive charge and end on negative charge and the magnetic lines of flux are closed lines.

32. What is Curl?

The curl is a closed line integral per unit area as the area shrinks to a point. It gives the circulation per unit area i.e. circulation density of a vector about a point at which the area is going to shrink. The curl also gives the direction, which is along the axis through a point at which curl is defined.

33. Give the relation between H and μ in tangential component.

The tangential component of H are continuous, while tangential component of B are discontinuous at the boundary, with the condition that the boundary is current free.

$$\frac{B_{\tan 1}}{B_{\tan 2}} = \frac{\mu_1}{\mu_2}.$$

34. Give the relation between H and μ in normal component.

The tangential component of H are not continuous at the boundary. The field strengths in two media are inversely proportional to their relative permeabilities.

$$\frac{H_{N1}}{H_{N2}} = \frac{\mu_2}{\mu_1}.$$

35. What is permeability?

In magnetostatics, the B and H are related to each other through the property of the

region in which current carrying conductor is placed. It is called permeability denoted as μ . It is the ability with which the current carrying conductor forces the magnetic flux through the region around it.

$$B = \mu H .$$

36. Distinguish between solenoid and toroid.

Solenoid is a cylindrically shaped coil consisting of a large number of closely spaced turns of insulated wire wound usually on a non magnetic frame. If a long slender solenoid is bent into the form of a ring and there by closed on itself it becomes a toroid.

37. Write the expression for inductance of a toroid?

$$L = \mu N^2 A / (2\pi R) \text{ H}$$

38. Write the expression for inductance of a solenoid?

$$L = \mu N^2 A / l \text{ H}$$

39. Write the expression for inductance of a coaxial cable?

$$L = \mu d / 2\pi \ln (b/a) \text{ H}$$

40. Describe what are the sources of electric field and magnetic field?

Stationary charges produce electric field that are constant in time, hence the term electrostatics. Moving charges produce magnetic fields hence the term magnetostatics.

41. Define current density.

Current density is defined as the current per unit area.

UNIT V
TWO MARKS

1. Define a wave.

If a physical phenomenon that occurs at one place at a given time is reproduced at other places at later times, the time delay being proportional to the space separation from the first location then the group of phenomena constitutes a wave.

2. Mention the properties of uniform plane wave.

i) At every point in space, the electric field E and magnetic field H are perpendicular to each other.

ii) The fields vary harmonically with time and at the same frequency everywhere in space.

3. Define intrinsic impedance or characteristic impedance.

It is the ratio of electric field to magnetic field. or It is the ratio of square root of permeability to permittivity of medium.

4. Give the characteristic impedance of free space.

$$377 \text{ ohms}$$

5. Define propagation constant.

Propagation constant is a complex number

$$\gamma = \alpha + j\beta$$

where γ is propagation constant

6. Define skin depth

It is defined as that depth in which the wave has been attenuated to $1/e$ or approximately 37% of its original value.

7. Define Poynting vector.

The pointing vector is defined as rate of flow of energy of a wave as it propagates.

$$\mathbf{P} = \mathbf{E} \times \mathbf{H}$$

8. State Poyntings Theorem.

The net power flowing out of a given volume is equal to the time rate of decrease of the the energy stored within the volume- conduction losses.

9. Give the difficulties in FDM.

FDM is difficult to apply for problems involving irregular boundaries and non homogeneous material properties.

10. Explain the steps in finite element method.

i) Discrimination of the solution region into elements.

- ii) Generation of equations for fields at each element
- iii) Assembly of all elements
- iv) Solution of the resulting system

11. State Maxwell's fourth equation.

The net magnetic flux emerging through any closed surface is zero.

12. State Maxwell's Third equation

The total electric displacement through the surface enclosing a volume is equal to the total charge within the volume.

13. State the principle of superposition of fields.

The total electric field at a point is the algebraic sum of the individual electric field at that point.

14. Define pointing vector.

The vector product of electric field intensity and magnetic field intensity at a point is a measure of the rate of energy flow per unit area at that point.

15. Give the formula to find potential at a point which is surrounded by four orthogonal points in FDM.

$$V_0 = \frac{1}{4}(V_1 + V_2 + V_3 + V_4)$$

16. Give the formula to find potential at a point which is surrounded by six orthogonal points in FDM.

$$V_0 = \frac{1}{6}(V_1 + V_2 + V_3 + V_4 + V_5 + V_6)$$

17. Define loss tangent.

Loss tangent is the ratio of the magnitude of conduction current density to displacement current density of the medium.

18. Define reflection and transmission coefficients.

Reflection coefficient is defined as the ratio of the magnitude of the reflected field to that of the incident field.

19. Define transmission coefficients.

Transmission coefficient is defined as the ratio of the magnitude of the transmitted field to that of incident field.

20. What will happen when the wave is incident obliquely over dielectric –dielectric boundary?

When a plane wave is incident obliquely on the surface of a perfect dielectric part of the energy is transmitted and part of it is reflected. But in this case the transmitted wave will be refracted, that is the direction of propagation is altered.

21. What is the fundamental difference between static electric and magnetic field

lines?

There is a fundamental difference between static electric and magnetic field lines. The tubes of electric flux originate and terminates on charges, whereas magnetic flux tubes are continuous.

22. What are uniform plane waves?

Electromagnetic waves which consist of electric and magnetic fields that are perpendicular to each other and to the direction of propagation and are uniform in plane perpendicular to the direction of propagation are known as uniform plane waves.

23. What is the significant feature of wave propagation in an imperfect dielectric ?

The only significant feature of wave propagation in an imperfect dielectric compared to that in a perfect dielectric is the attenuation undergone by the wave.

24. What is the major drawback of finite difference method?

The major drawback of finite difference method is its inability to handle curved boundaries accurately.

25. What is method of images?

The replacement of the actual problem with boundaries by an enlarged region or with image charges but no boundaries is called the method of images.

26. When is method of images used?

Method of images is used in solving problems of one or more point charges in the presence of boundary surfaces.

27. Define power density.

The power density is defined as the ratio of power to unit area.

$$\text{Power density} = \text{power/unit area.}$$

28. What is called wave velocity?

The velocity of propagation is called as wave velocity. It is denoted as v .

$$v = \frac{1}{\sqrt{\mu\epsilon}}$$

For free space it is denoted by c and its value is $3 \times 10^8 \text{ m/s}$.

29. What is called as intrinsic impedance?

The ratio of amplitudes of E and H of the waves in either direction is called intrinsic impedance of the material in which wave is travelling. It is denoted by Z_0 .

30. Why dielectric medium is lossless dielectric.

For perfect dielectric medium, both the fields E and H are in phase. Hence there is no attenuation. Hence there is no loss.

31. What is mean by lossy dielectric?

The presence of attenuation indicates there is a loss in the medium. Hence such medium is called as lossy dielectric.

32. What is mean by skin depth?

The distance through which the amplitude of the travelling wave decreases to 37% of the original amplitude is called skin depth or depth of penetration.

33. What is called skin effect?

For the frequencies in the microwave range, the skin depth or depth of penetration is very small for good conductors and all the fields and currents may be considered as confined to a thin layer near the surface of the conductor. This thin layer is nothing but the skin of the conductor and hence it is called skin effect.

34. What is Normal Incidence?

When a uniform plane wave incidences normally to the boundary between the media, then it is known as normal incidence.

35. What is normal Incidence?

When a uniform plane wave incidences obliquely to the boundary between the media, then it is known as normal incidence.

36. What are Waves?

Basically the waves are means of transporting energy or information from source to destination. Also a wave is function of both space and time. The typical ex of EM waves are radio waves, TV signals, radar beams.

37. Give Wave equation in differential form.

$$\frac{\partial^2 E_x}{\partial t^2} = v^2 \frac{\partial^2 E_x}{\partial z^2}.$$

38. What is called attenuation constant?

When a wave propagates in the medium, it gets attenuated. The amplitude of the signal reduces. This is represented by attenuation constant. It is measured in neper per meter (NP/m). But practically it is expressed in decibel (dB).

39. What is phase constant?

When a wave propagates, phase change also takes place. Such a phase change is expressed by a phase constant β . It is measured in radian per meter (rad/m).

40. Define standing wave ratio.

The standing wave ratio is defined as the ratio of maximum to minimum amplitudes of voltage.

$$S = \frac{E_{1s \max}}{E_{1s \min}}.$$

41. How voltage maxima and minima are separated?

In general voltage minima are separated by one half wavelength. Also the voltage maxima are also separated by one half wave length.

42. What is the condition for perfect dielectric?

For perfect dielectric, the conductivity is zero and hence the loss of the system is also zero.

43. What is the condition for practical dielectric?

For practical dielectric, there is some conductivity, that is its value is not zero and hence there is some loss in practical dielectric but its value is very small.

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